MILITARY SPECIFICATION

PERISCOPE, TANK: XM50

1. SCOPE

1.1 This specification covers one type of gunner's periscope which provides a unity-power viewing, an eight-power daylight, and a ten-power passive (low-light level) sighting system. The periscope is equipped with a reticle image projector which projects the image of a reticle pattern into the sighting systems. The projector also electro-mechanically positions the reticle pattern, in elevation and deflection, to coincide with correction data received from a computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-150	Photographic Lenses
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10539750	Periscope, Tank: XM50
F8565556	Pressure Tester-Two Station

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PACKAGING DATA SHEET

MIL-P-14232/P10539750 Packaging of Periscope, Tank: XM50

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Index of item requirements.- The Tank Periscope: XM50 and sub-assemblies thereof, tabulated in Table I, shall be capable of meeting the applicable requirements listed.

TABLE I

Item	Requirement
1. Head assembly	3.2 thru 3.9 and 3.10
2. Body assembly	3.2 thru 3.9 and 3.11
3. Reticle image projector	3.2 thru 3.9 and 3.12
4. Periscope, Tank: XM50	3.2 thru 3.9 and 3.13

3.2 First article testing.- Requirements for the submission of first article samples for testing shall be as specified in the contract.

3.3 Fabrication.- The periscope shall be manufactured in accordance with Drawing F10539750. The periscope subassemblies, tabulated in Table I, shall be manufactured in accordance with the applicable subassembly drawings pertaining to Drawing F10539750.

3.4 Temperature conditions.- Except as specified in 3.7 and 3.8, the requirements hereafter shall be met at an ambient temperature between +70° and +80° Fahrenheit (F).

3.5 General specification.- The following requirements of MIL-F-13926 shall be met wherever applicable.

(a) Dimensions and tolerances.
(b) Inorganic protective surface finishes.
(c) Electrical and electronic assemblies.
(d) Part identification and marking.
(e) Soldering electrical connections.

3.6 Workmanship.- The workmanship requirements of MIL-F-13926 shall be adhered to. Exterior protective surface finishes shall be free of voids which may affect the protective qualities of the finish or appearance. Optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

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3.7 Storage temperatures.- There shall be no evidence of physical failures such as glass breakage, bonding failure, or casting failure due to storage at temperatures ranging from -65° to +160°F.

3.8 Extreme operating temperatures.- Sustained operation (see 3.10.9, 3.11.4 and 3.12.7.3) shall be met at extreme temperatures ranging from -40° to +150°F (where applicable).

3.9 Vibration.- The equipment shall show no evidence of physical failure due to being vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. There shall be no failure or flicker of light emission from the reticle projector lamp and image tube during vibration. Subsequent to vibration the equipment shall be capable of meeting all requirements of this specification.

3.10 Head assembly.

3.10.1 Orientation.- With the mounting surface of the head assembly that mates to the body assembly in a horizontal plane, and with the back face of the coupling set at 67°30' to the head assembly horizontal mounting surface, an incident line of sight perpendicular to the head assembly mounting surface shall be reflected 1°30' plus or minus 0.2 mil below the horizontal plane. This line of sight shall be perpendicular in azimuth within plus or minus 0.2 mil to a vertical plane, derived from the axis of the two 3/16 inch diameter locating pins at the front of the head assembly. These pins locate the head assembly to the body assembly.

3.10.2 Excursion.- Angular movement of the coupling shall be at least 40 degrees within the limit stops and shall elevate the line of sight at least 22° and depress it at least 18° from the position established in 3.10.1.

3.10.3 Elevation accuracy.- Angular changes at the input coupling shall cause the same change in the line of sight within 0.2 mil, including backlash, over the total excursion of coupling travel.

3.10.4 Backlash.- Backlash between the input coupling and the head mirror shall not exceed 0.05 mil at any position over the full excursion of the coupling movement.

3.10.5 Plumb travel.- With the head positioned as in 3.10.1, and with the line of sight horizontal and in coincidence with a plumb line, the line of sight shall not deviate from the plumb line in excess of 0.25 mil when elevated 20 degrees above the horizontal and depressed 18 degrees below the horizontal line of sight.

3.10.6 Resolution.- The resolution shall be 2.0 seconds of arc or less, using a telescope of 80 to 120 power with an entrance pupil of at least seven inches, at one focus setting.

3.10.7 Luminous transmittance.- Luminous transmittance shall be not less than 90 percent.

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3.10.8 Transmittance at 0.85 micron.- Transmittance at 0.85 micron shall be not less than 90 percent.

3.10.9 Operability.- The maximum torque, as applied at the coupling required to rotate the head mirror through its full excursion (stop to stop) shall not exceed 7 pound-inches between +70° and +80°F, and 12 pound-inches at temperatures ranging from -40° to +150°F.

3.10.10 Cleanliness.- The head assembly shall meet the cleanliness requirements of MIL-O-13830. There shall be not more than four particles of dirt visible on the optical surface in the interior of the head assembly. The area of each particle shall not be in excess of an area equivalent to a size ten dig. There shall be no foreign matter which interferes with observation or adversely affects image quality.

3.10.11 Sealing.- The head assembly shall show no evidence of leakage when subjected to an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) over a period of at least 5 minutes.

3.11 Body assembly.- The body assembly shall meet the requirements of 3.11.1 through 3.11.4.3.2 with the mounting surface of the body assembly, that mates to the guard and shield assembly in a horizontal plane.

3.11.1 Unity-power system.

3.11.1.1 Resolution.- Resolution shall be one minute of arc or less when viewed through any area of the exit window with a 3-power telescope having at least a 1/2 inch aperture.

3.11.1.2 Mirror repeatability.- The line of sight shall return to an established position, within plus or minus one degree, when the mirror introduction lever is actuated to move the mirror of the unity power system out of and back into the operational position.

3.11.2 Daylight system.

3.11.2.1 Resolution.- The resolution on the optical axis shall be 7 seconds of arc or less as viewed with an auxiliary telescope of at least 3 power at one focus setting.

3.11.2.2 Image tilt.- The image of a target line originally perpendicular to the plane of the periscope locating keyway shall emerge perpendicular to the horizontal locating surface of the periscope within one degree.

3.11.2.3 Eyepiece focus.- Eyepiece focus shall be adjustable over the range of plus and minus 4 diopters as read on the eyepiece scale. The zero setting of the diopter scale shall indicate zero diopter within 1/4 diopter when the image of an infinity target is in sharp focus.

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3.11.2.4 Eyepiece backlash.- Backlash in the eyepiece focus (diopter movement) shall not exceed 1/8 diopter.

3.11.3 Passive system.

3.11.3.1 Contrast transfer (with image tube removed).

3.11.3.1.1 Objective system.- The contrast transfer for vertical and horizontal lines shall be not less than the values specified in Table II.

TABLE II

Angular spacing (in line pairs/milliradian)	Contrast transfer =% Vertical lines	Horizontal line
On axis		
0.58	0.95	0.95
1.732	0.91	0.91
4.62	0.82	0.82
6.92	0.73	0.73
9.23	0.64	0.64
Off axis 0.3 inch		
0.58	0.93	0.93
1.732	0.88	0.88
4.62	0.75	0.75
6.92	0.65	0.65
9.23	0.50	0.50

NOTE: The values (for the axis specified) shall be met with 2870° Kelvin (K) radiance, modulated at the spatial frequency (angular spacing) indicated, incident to the entrance window and the contrast values obtained at the objective system focal plane. "On axis" is defined as the geometric axis of Housing Assembly 10539615 (Drawing C10539615). "Off axis" is defined as displacement, conforming to the value specified, from on axis.

3.11.3.1.2 Eyepiece and erecting system.- The contrast transfer for vertical and horizontal lines shall be not less than the values specified in Table III.

TABLE III

Angular spacing (in line pairs/milliradian)	Contrast transfer =% Vertical lines	Horizontal lines
On axis		
0.58	0.98	0.98
1.732	0.96	0.96
4.62	0.84	0.84
6.96	0.75	0.75
9.23	0.66	0.66

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TABLE III (Cont'd)

Off axis 0.3 inch

Angular spacing (in line pairs/milliradian)	Contrast transfer =% Vertical lines	Horizontal lines
0.58	0.95	0.95
1.732	0.92	0.92
4.62	0.70	0.70
6.92	0.70	0.70
9.23	0.55	0.55

3.11.3.2 Radiant transmittance (with image tube removed).

NOTE: Radiant transmittance requirement shall be met with a source of radiance at the focal plane of the system specified.

3.11.3.2.1 Objective system.- a. Without filters (clear position). Luminous transmittance shall be not less than 58%, when observed through any area of the entrance pupil, with the filter in the clear position.

b. With filters.- There shall be a full field of view, with no indication of cut-off, when the filter assembly is in detent at each of the three viewing positions. The ratio of luminous transmittance, for the filter positions specified to that obtained with filter in clear position, shall be as indicated in Table IV.

TABLE IV

Filter position	Ratio in percentage Maximum	Minimum
Medium	3.7	2.5
Dark	0.127	0.08

3.11.3.2.2 Transmittance at 0.85 micron. a. Without filters (clear position).- Transmittance at 0.85 micron shall be not less than 58% with filter in clear position.

b. With filters.- The ratio of luminous transmission at 0.85 micron, for the filter positions specified to that obtained with filter in clear position at 0.85 micron, shall be as indicated in Table V.

TABLE V

Filter position	Ratio in percentage Maximum	Minimum
Medium	3.7	2.5
Dark	0.127	0.08

3.11.3.3 Eyepiece and erecting system.- Luminous transmittance shall be not less than 60% when observed through the exit pupil.

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3.11.3.4 Resolution (with image tube removed).

3.11.3.4.1 Objective system.- With the specified line pairs per millimeter position at, and projected from, the focal plane of the objective system the following shall be met:

a. Resolve 200 line pairs per millimeter on axis.
b. Resolve 200 line pairs per millimeter off axis (0.35 inch).

3.11.3.4.1.1 Objective T-number.

a. On axis T-number shall be not more than 2.74 as observed on axis for the objective system.
b. Off axis T-number shall be not more than 3.20 as observed off axis 0.35 inch.

NOTE: The source of radiance shall be at the focal plane of the objective system. The T-number is defined as the $\frac{f}{D}$ (ratio of focal length to equivalent diameter) of an ideal circular lens having no observation or transmission loss, and has the equivalent focal length of the system in question and delivers the same irradiance at the focal plane as the system in question.

3.11.3.5 Resolution.- With the image tube, oscillator, and voltage regulator installed and energized, the requirements of 3.11.3.5 to 3.11.3.11.1 inclusive shall be met. The passive system shall meet the requirements of Table VI, for the axis specified, when observing a target having the background luminance in foot-lamberts (FT-L) and the contrast specified.

TABLE VI

Resolution		Resolution target	
On axis	Off axis 2.0 degree	Background luminance	Contrast
$0.78R_T$	$0.65R_T$	1.5×10^{-2} FT-L	High 1.0 or greater
$0.17R_T$		6×10^{-5} FT-1	Low 0.20 $\pm$ 0.5

NOTE: Contrast, for the purpose of this requirement is defined as the difference in density between dark lines and the light background. R_T is the minimum on axis resolution requirement for the image intensifier tube. The detailed requirements for the image intensifier tube are specified in Purchase Description for Image Intensifier Tube Assembly (25 millimeter), Type 8586, dated 8 July 1964. Only the title of this purchase description is unclassified. The purchase description is classified Confidential.

3.11.3.6 Contrast transfer (with image tube installed and energized).- The contrast transfer for vertical and horizontal lines shall be not less than the values specified in Table VII.

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TABLE VII

Angular spacing (in line pairs/milliradian)	Contrast transfer Vertical lines	Horizontal lines
	On axis	
0.58	$0.90U_T$	$0.90U_T$
1.73	$0.85U_T$	$0.85U_T$
4.62	$0.65U_T$	$0.65U_T$
	Off axis 2.0 degrees	
0.58	$0.85U_T$	$0.85U_T$
1.73	$0.80U_T$	$0.80U_T$
4.62	$0.55U_T$	$0.55U_T$

NOTE: The values for axis specified shall be met with 2870°K radiance, modulated at the spatial frequency (angular spacing) indicated, incident to the entrance aperture of the periscope. U_T is the contrast transfer value obtained from the image intensifier tube assembly installed in the periscope being tested. The U_T values for the image intensifier tube assembly shall be converted to line pairs per milliradian.

3.11.3.7 Luminance gain.- The luminance gain shall be not less than 0.02 G_T.

NOTE: G_T is the luminance gain value obtained for the image tube intensifier assembly installed in the periscope being tested.

3.11.3.8 Stray light luminance.- Stray light luminance shall not exceed 2.0 percent when the assembly is set for infinity focus.

NOTE: Stray light luminance shall be the contribution over the image of a black disk, subtending a solid angle equal to the field of view, in a white surround subtending a solid angle of not less than 120 degrees.

3.11.3.9 Field of view.- The field of view shall be not less than 90 milliradians.

3.11.3.10 Projected reticle pattern (passive system).

3.11.3.10.1 Definition and visibility - clear filter.- The image of an aligned and projected reticle pattern shall be completely defined and visible to the unaided eye when a clear field having a luminance of 10^{-2} foot lamberts is presented to the entrance window.

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3.11.3.10.2 Definition and visibility - clear, medium, and dark filters.- The image of the reticle pattern shall be completely defined and visible, and of approximately equal brightness to the unaided eye, when the entrance window is covered to prevent entrance of light the reticle intensity knob is positioned for one best setting, and the clear, medium, and dark filters are positioned in sequence in the optical train.

3.11.3.11 Image steadiness.

3.11.3.11.1 Flicker.- There shall be no evidence of flicker or movement of the image subsequent to energizing the tube and allowing a 15-second warm-up period.

3.11.3.11.2 Sparking.- There shall be no evidence of sparking or arcing at any time.

3.11.3.12 Objective focus.- Rotation of the objective focus control knob shall bring targets located from 50 meters distant to infinity into sharp focus.

3.11.3.12.1 Diopter change.- The diopter setting shall not change by more than 0.25 diopter as read on the diopter scale.

3.11.4 Operability.

3.11.4.1 Diopter movement.- The running torque required to rotate the diopter adjustment shall not exceed 7 pound-inches between +70° and +80°F, and 12 pound-inches at temperatures ranging from -40° to +150°F.

3.11.4.2 Passive, daylight, and unity power introduction levers.- The passive, daylight, and the unity power introduction levers shall be mechanically operable and lock in detent at each position (in and out) between +70° and +80°F and at temperatures ranging from -40° to +150°F. The force required to disengage the detents shall not exceed 5 pounds between +70° and +80°F, or 8 pounds at temperatures ranging from -40° to +150°F. The detent shall hold the filter assembly in a secured position at each of the four settings.

3.11.4.3 Micro-switch.- The micro-switch shall open when the filter assembly is at the "OFF" (blacked-out) position, and shall close, and remain closed, at the other three positions when checked to the requirements of 3.11.4.3.2.

3.11.4.3.1 Rheostat control.- The running torque required to rotate the rheostat control knob through its entire excursion shall not exceed 24 ounce-inches between +70° and +80°F, and 35 ounce-inches at temperatures ranging from -40° to +150°F.

3.11.4.3.2 Objective focus control.- The running torque required to rotate the objective focus control knob through its entire excursion shall not exceed 10 pound-inches between +70° and +80°F, and 15 pound-inches at temperatures ranging from -40° to +150°F.

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3.11.5 Cleanliness.- The body assembly shall meet the cleanliness requirements of MIL-O-13830. There shall be no foreign matter within the body assembly which interferes with observation or adversely affects image quality.

3.11.5.1 Unity power system.- There shall be no more than six particles of dirt visible on the optical surfaces in the interior of the unity power system. The area of each particle shall not be in excess of an area equivalent to a size twenty dig.

3.11.5.2 Passive system.- With the image intensifier tube energized, there shall be no more than three particles of dirt visible on the optical surfaces. The area of each particle shall not be in excess of an area equivalent to a size ten dig.

3.11.5.3 Daylight system.- There shall be no more than four particles of dirt visible on the optical surfaces. The area of each particle shall not be in excess of an area equivalent to a size ten dig.

3.11.6 Sealing.- The body assembly shall show no evidence of leakage when subjected to an internal pressure of 5 $\pm$ 0.10 pounds per square inch gage (psig) over a period of at least 5 minutes.

3.11.7 Interior atmosphere.- The interior atmosphere of the accepted body assembly shall contain nitrogen having a dewpoint at least as low as -25°F. The internal pressure of the accepted unit shall be between 0.10 and 0.25 psig.

3.12 Reticle image projector. (as viewed through periscope eyepiece)

3.12.1 Orientation.- The reticle image projector shall be positioned with the mounting surface to the periscope body in a vertical plane. The geometric axis of the circular housing for the objective (projection) lens and the axis of the alignment pin at the lower part of the projector housing shall be in coincidence with a plumb line within five minutes of arc.

3.12.2 Electrical-optical accuracy.- The ratio of reticle movement shall be 0.4 volt per one mil, plus or minus 0.1 mil, reticle movement over the entire excursions specified in 3.12.2.1 and 3.12.2.2.

3.12.2.1 Horizontal excursion.- With the boresight cross in coincidence with a vertical target line passing through the geometric axis of the field of view, the boresight cross shall move through an excursion of at least 28 mils to the left and 28 mils to the right of the vertical target line.

3.12.2.2 Vertical excursion.- With the boresight cross of the reticle positioned 26.67 mils above the geometric axis of the field of view and in coincidence with a vertical target line passing through the geometric axis, the boresight cross shall travel in depression through an excursion of at least 55 mils.

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3.12.2.3 Horizontal reticle travel.- Starting in coincidence with a horizontal target line, the boresight cross shall not deviate from the target line in excess of plus or minus 0.05 mil (total deviation 0.10 mil) over the excursion specified in 3.12.2.1.

3.12.2.4 Vertical reticle travel.- Starting in coincidence with a vertical target line, the boresight cross shall not deviate from the target line in excess of plus or minus 0.05 mil (total deviation 0.10 mil) over the excursion specified in 3.12.2.2.

3.12.2.5 Lateral displacement.- Change in direction of reticle travel shall not cause the reticle to shift by more than 0.15 mil in a direction perpendicular to the intended movement of the reticle, including backlash.

3.12.2.6 Backlash.- Backlash in reticle movement shall not exceed 0.075 mil.

3.12.2.7 Stability.- The reticle shall remain stable at any established position. Jump, creep, or shift shall not exceed 0.05 mil in elevation and deflection.

3.12.3 Reticle focus.- The reticle shall be in sharp focus between zero and minus 3/4 diopter.

3.12.4 Reticle accuracy.- The angular value between the boresight cross and the maximum graduations of the reticle pattern shall be within 3.0 percent of the values specified on the reticle drawing.

3.12.5 Parallax.- Parallax between the image of the complete reticle pattern and the image in an auxiliary telescope, adjusted for zero parallax at infinity, shall not exceed 0.1 mil.

3.12.6 Parallelism of reticle and image.- With the reticle projector positioned as in 3.12.1, the total spread of the reticle pattern through the boresight cross shall be parallel to the image of a horizontal target line of an auxiliary telescope within 30 minutes of arc when the boresight cross is in coincidence with a representative plumb line that passes through the geometric axis of the field of view.

3.12.7 Electrical phasing.- When the circuit is energized in accordance with Schematic Drawing F10540243, the reticle shall move as required in 3.12.7.1 and 3.12.7.2.

3.12.7.1 Horizontal movement.- The reticle shall move to the right when the applied voltage at pin 5 is in phase with the applied voltage at pin 2. The reticle shall move to the left when this voltage is 180 degrees out-of-phase.

3.12.7.2 Vertical movement.- The reticle shall move in depression when the applied voltage at pin 9 is in phase with the applied voltage at pin 2. The reticle shall move in elevation when this voltage is 180 degrees out-of-phase.

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3.12.7.3 Slewing speed.- The reticle shall travel through its total excursion of horizontal and vertical travel in a time cycle of 6 seconds or less at temperatures ranging from -40° to +150°F, and 3 seconds or less between +70° and +80°F.

3.12.8 Illumination.- A 24-volt power source, when applied to pins 1B and 6 (ground) of connection J, shall illuminate the reticle pattern.

3.12.9 Dielectric strength.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the projector are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than 5 seconds.

3.12.10 Insulation resistance.- All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts direct-current (vdc) applied.

3.12.11 Cleanliness.- There shall be no evidence of dirt to obscure any portion of the projected markings of the reticle pattern when the reticle lamp is energized from minimum to maximum intensity.

3.12.12 Sealing.- The reticle image projector shall show no evidence of leakage when subjected to an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) over a period of at least 5 minutes.

3.12.13 Interior atmosphere.- The interior atmosphere of the accepted reticle image projector shall contain nitrogen having a dewpoint at least as low as -25°F. The internal pressure of the accepted unit shall be between 0.10 and 0.25 psig.

3.13 Periscope: XM50.- The periscope shall consist of the Head Assembly:, Body Assembly:, and Reticle Projector.

3.13.1 Orientation.- With the mounting surface of the body assembly mounted to the guard and shield assembly in a horizontal plane (the surface that mounts the complete periscope in its operating position), the following requirements shall be met:

Requirement	Paragraph
a. Excursion	3.10.2
b. Elevation accuracy	3.10.3
c. Backlash	3.10.4
d. Mirror repeatability	3.11.1.2
e. Contrast transfer	3.11.3.1
f. Radiant transmittance	3.11.3.2
g. Resolution	3.11.3.4
h. Field of view	3.11.3.9
i. Image steadiness	3.11.3.11
j. Horizontal excursion	3.12.2.1
k. Vertical excursion	3.12.2.2
l. Horizontal reticle travel	3.12.2.3
m. Vertical reticle travel	3.12.2.4

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3.13.2 Parallax.- Parallax at the optical axes through the daylight system shall not exceed 0.10 mil, when viewing a target 1000 meters distant, plus or minus 100 meters.

3.13.3 Parallelism of reticle image.- The image of the total spread of the horizontal reticle line (over length of lines and spaces) shall be parallel to the image of a horizontal target line within 10 minutes of arc.

3.13.4 Repeatability.- Actuation of the passive, daylight introduction lever to go from passive to daylight or vice-versa, shall not cause a shift in the image in excess of one degree as read at the eyepiece.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3,4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

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4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables VIII through XVI herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE VIII - CLASSIFICATION OF DEFECTS

HEAD ASSEMBLY

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 0.65% defective			
101.	Vibration	3.9	4.6.3.1
102.	Line of sight orientation	3.10.1	4.6.4.1.1
103.	Excursion	3.10.2	4.6.4.1.2
104.	Elevation accuracy	3.10.3	4.6.4.1.3
105.	Plumb travel	3.10.5	4.6.4.1.4
106.	Backlash	3.10.4	4.6.4.1.5
107.	Luminous transmittance	3.10.7	4.6.4.1.6
108.	Transmittance at 0.85 micron	3.10.8	4.6.4.1.7
109.	Resolution	3.10.6	4.6.4.2
110.	Operability	3.10.9	4.6.4.3
111.	Cleanliness	3.10.10	4.6.4.4
112.	Sealing	3.10.11	4.6.4.5
MINOR: NONE			

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TABLE IX - CLASSIFICATION OF DEFECTS

BODY ASSEMBLY

CLASS CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 0.65% defective		
113. Vibration	3.9	4.6.3.2
114. Contrast transfer, on axis, objective system	3.11.3.1.1	4.6.5.1.1
115. Contrast transfer, on axis, eyepiece and erecting system	3.11.3.1.2	4.6.5.1.3
116. Radiant transmittance, objective system, without filters	3.11.3.2.1a	4.6.5.1.4
117. Radiant transmittance, objective system, with filters	3.11.3.2.1b	4.6.5.1.5
118. Transmittance at 0.85 micron, objective system, without filters	3.11.3.2.2a	4.6.5.1.6
119. Transmittance at 0.85 micron, objective system, with filters	3.11.3.2.2b	4.6.5.1.7
120. Eyepiece and erecting system, luminous transmittance	3.11.3.3	4.6.5.1.8
121. Resolution, on axis, objective system	3.11.3.4.1a	4.6.5.1.9
122. Resolution, unity power system	3.11.1.1	4.6.5.1.13
123. Resolution, daylight system	3.11.2.1	4.6.5.1.14
124. Resolution, passive system, on axis	3.11.3.5	4.6.5.1.16
125. Contrast transfer, on axis	3.11.3.6	4.6.5.1.18
126. Luminance gain	3.11.3.7	4.6.5.1.20
127. Field of view	3.11.3.9	4.6.5.1.22
128. Definition and visibility, projected reticle pattern, clear filter	3.11.3.10.1	4.6.5.1.23
129. Definition and visibility, projected reticle pattern, clear, medium and dark filters	3.11.3.10.2	4.6.5.1.24
130. Flicker, image steadiness	3.11.3.11.1	4.6.5.1.25
131. Sparking, image steadiness	3.11.3.11.2	4.6.5.1.26
132. Cleanliness, passive system	3.11.5.2	4.6.5.1.27
133. Cleanliness, unity power system	3.11.5.1	4.6.5.1.28
134. Cleanliness, daylight system	3.11.5.3	4.6.5.1.29
135. Mirror repeatability, unity power system	3.11.1.2	4.6.5.2.1
136. Image tilt, daylight system	3.11.2.2	4.6.5.2.2
137. Eyepiece focus, daylight system	3.11.2.3	4.6.5.2.3
138. Eyepiece backlash, daylight system	3.11.2.4	4.6.5.2.4

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TABLE IX - CLASSIFICATION OF DEFECTS (Cont'd)

		REQUIREMENT	TEST PROCEDURES
139.	Objective focus	3.11.3.12	4.6.5.2.5
140.	Diopter change	3.11.3.12.1	4.6.5.2.6
141.	Operability, diopter movement	3.11.4.1	4.6.5.3
142.	Operability, passive, daylight and unity power introduction levers	3.11.4.2	4.6.5.4
143.	Operability, micro-switch	3.11.4.3	4.6.5.5
144.	Operability, rheostat control	3.11.4.3.1	4.6.5.6
145.	Operability, objective focus control	3.11.4.3.2	4.6.5.7
146.	Interior atmosphere	3.11.7	4.6.5.8
147.	Sealing	3.11.6	4.6.5.9

MINOR: NONE

TABLE X - CLASSIFICATION OF DEFECTS

RETICLE IMAGE PROJECTOR

CLASS CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 0.65% defective			
148.	Vibration	3.9	4.6.3.3
149.	Horizontal excursion	3.12.2.1	4.6.6.1.2
150.	Vertical excursion	3.12.2.2	4.6.6.1.3
151.	Horizontal reticle travel	3.12.2.3	4.6.6.1.4
152.	Vertical reticle travel	3.12.2.4	4.6.6.1.5
153.	Lateral displacement	3.12.2.5	4.6.6.1.6
154.	Backlash	3.12.2.6	4.6.6.1.7
155.	Stability	3.12.2.7	4.6.6.1.8
156.	Reticle focus	3.12.3	4.6.6.1.9
157.	Reticle accuracy	3.12.4	4.6.6.1.10
158.	Parallax	3.12.5	4.6.6.1.11
159.	Parallelism of reticle and image	3.12.6	4.6.6.1.12
160.	Horizontal movement	3.12.7.1	4.6.6.1.13.1
161.	Vertical movement	3.12.7.2	4.6.6.1.13.2
162.	Slewing speed	3.12.7.3	4.6.6.1.13.3
163.	Illumination	3.12.8	4.6.6.1.14
164.	Dielectric strength	3.12.9	4.6.6.2
165.	Insulation resistance	3.12.10	4.6.6.3
166.	Cleanliness	3.12.11	4.6.6.4
167.	Sealing	3.12.2	4.6.6.5
168.	Interior atmosphere	3.12.13	4.6.6.6
MINOR: NONE			

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TABLE XI - CLASSIFICATION OF DEFECTS

PERISCOPE TANK: XM50

CLASS CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 0.65% defective			
169.	Vibration	3.9	4.6.3.4
170.	Contrast transfer	3.13.1e	4.6.7.1.1
171.	Radiant transmittance	3.13.1f	4.6.7.1.2
172.	Resolution	3.13.1g	4.6.7.1.3
173.	Field of view	3.13.1h	4.6.7.1.5
174.	Image steadiness	3.13.1i	4.6.7.1.6
175.	Excursion	3.13.1a	4.6.7.2.1
176.	Elevation accuracy	3.13.1b	4.6.7.2.2
177.	Backlash	3.13.1c	4.6.7.2.3
178.	Mirror repeatability	3.13.1d	4.6.7.2.4
179.	Horizontal excursion	3.13.1j	4.6.7.2.5
180.	Vertical excursion	3.13.1k	4.6.7.2.6
181.	Horizontal reticle travel	3.13.1L	4.6.7.2.7
182.	Vertical reticle travel	3.13.1m	4.6.7.2.8
183.	Parallax, daylight system	3.13.2	4.6.7.2.9
184.	Parallelism of reticle image	3.13.3	4.6.7.2.10
185.	Repeatability	3.13.4	4.6.7.2.11
MINOR: NONE			

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One sample of each item procured under this specification, shall be selected at random as a special sampling from each 100 produced or from each month's production whichever occurs first. The sample shall meet the requirements and tests in Table XII.

TABLE XII

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.3	Applicable Drawings - visual
302.	General specification	3.5	MIL-F-13926-Visual
303.	Workmanship	3.6	Visual - tactile

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4.4.2 Environmental.

4.4.2.1 Head assembly.- Three head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table VIII and shall then meet the requirements and tests in Table XIII.

TABLE XIII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
304. Storage temperatures (-65° to +160°F)	3.7	4.6.1
305. Extreme operating temperatures (-40° to +150°F)	3.8	4.6.2

4.4.2.2 Body assembly.- Three body assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table IX and shall then meet the requirements and tests in Table XIV.

TABLE XIV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
306. Storage temperatures (-65° to +160°F)	3.7	4.6.1
307. Extreme operating temperatures (-40° to +150°F)	3.8	4.6.2
308. Vibration	3.9	4.6.3.3
309. Contrast transfer, off-axis, objective system (+60° to 90°F)	3.11.3.1.1	4.6.5.1.2
310. Contrast transfer, off-axis, eyepiece and erecting system (+60° to 90°F)	3.11.3.1.2	4.6.5.1.3
311. Resolution, off-axis, objective system (+60° to 90°F)	3.11.3.4.1b	4.6.5.1.10
312. Objective T-number, on-axis (+60° to 90°F)	3.11.3.4.1.1a	4.6.5.1.11
313. Objective T-number, off-axis (+60° to 90°F)	3.11.3.4.1.1b	4.6.5.1.12
314. Resolution, passive system, off-axis (+60° to 90°F)	3.11.3.5	4.6.5.1.17
315. Contrast transfer, off-axis (+60° to 90°F)	3.11.3.6	4.6.5.1.19
316. Stray light luminance (+60° to 90°F)	3.11.3.8	4.6.5.1.21

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4.4.2.3 Reticle image projector.- Three reticle image projectors shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table X and shall then meet the requirements and tests in Table XV.

TABLE XV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
317. Storage temperatures (-65° to +160°F)	3.7	4.6.1
318. Extreme operating temperatures (-40° and +150°F)	3.8	4.6.2

4.4.2.4 Periscope, Tank, XM50.- Three periscopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table XI and shall then meet the requirements and tests in Table XVI.

TABLE XVI

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
319. Storage temperatures (-65° to +160°F)	3.7	4.6.1
320. Extreme operating temperatures (-40° and +150°F)	3.8	4.6.2
321. Contrast transfer, off-axis, objective system (+60° to 90°F)	3.13.1e	4.6.5.1.2, 4.6.7.1.2
322. Contrast transfer, off-axis, eyepiece and erecting system (+60° to 90°F)	3.13.1e	4.6.5.1.3, 4.6.7.1.2
323. Resolution, off-axis, objective system (+60° to 90°F)	3.13.1g	4.6.7.1.4, 4.6.5.1.10
324. Objective T-number, on-axis (+60° to 90°F)	3.13.1g	4.6.7.1.4, 4.6.5.1.11
325. Objective T-number, off-axis (+60° to 90°F)	3.13.1g	4.6.7.1.4, 4.6.5.1.12

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special testing equipment to test sealing of each item procured under this specification shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester - Two Station.

4.5.2.3.2 Orientation - head assembly.- The special testing equipment utilized to mount and orient the head assembly for the accuracy tests shall conform to the following requirements.

a. The testing equipment shall be capable of mounting the head assembly in its normal operating position, as depicted on Drawing F10539750. Adjustments shall be provided on the testing equipment that shall permit orientation of Datum Plane "A", as shown on Sheet 2 of Drawing F10539748, in a horizontal plane within 10 seconds.

b. A device that is capable of positioning and solidly arresting the back face of the coupling, 67 degrees 30 minutes within 10 seconds with respect to Datum Plane "A", shall be a part of the test equipment. Subsequent to the above positioning, this device shall be capable of articulation of the back face of the coupling to any angular position between its travel stops.

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c. A target projector collimator having a clear aperture of at least 2.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned (see Figure 1) so that the optical axis of the collimator is in alinement with the line of sight emitting from the central aperture (geometric axis) of the head assembly. The target projector collimator shall have an appropriate calibrated target reticle pattern, containing a boresight mark 1 degree 30 minutes below the geometric axis of the reticle, for projection into the optical system of the head assembly. The vertical reticle line of the target test reticle pattern shall be adjusted plumb. The target projector collimator test reticle shall be within the accuracies indicated for the following requirements:

Elevation accuracy	5 seconds of arc
Plumb travel	"
Backlash	"
Excursion, elevation	10 seconds of arc
Excursion, depression	"

The optical axis of the target projector collimator shall be alined parallel with respect to the optical axis of the viewing autocollimator and autocollimated to a vertical plane containing the longitudinal axis of the 2 locating pins (MS9390-30) of the head assembly (see Sheet 1 of Drawing F10539749)within 5 seconds.

d. A penta prism having a deviation of ±1 second (calibrated) shall be a part of the test equipment. The penta prism shall be positioned (see Figure 1) so that the optical axis of the penta prism is in alinement with the line of sight emitting from the central aperture (geometric axis) of the head assembly.

e. A viewing autocollimator capable of measuring any line of sight deviations, in the "X" and "Y" coordinates in seconds of arc shall be a part of the test equipment. The viewing autocollimator shall be positioned at the penta prism (see Figure 1) so that the optical axis of the viewing autocollimator is in alinement with the line of sight emitting from the central aperture (geometric axis) of the head assembly. The optical axis of the viewing collimator shall be autocollimated through the penta prism, to Datum Plane "A" within 5 seconds.

f. All set-up and calibration equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and shall become part of the testing equipment.

4.5.2.3.3 Orientation - body assembly.- The special testing equipment utilized to mount and orient the body assembly for the accuracy tests shall conform to the following requirements.

a. The testing equipment shall be capable of mounting the body assembly in its normal operating position, as depicted on Drawing F10539750.

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Adjustments shall be provided on the testing equipment that shall permit orientation of Datum Plane "A", shown on Sheet 1 of Drawing F10539736; in a horizontal plane within 10 seconds. The necessary mounting features (keys and locating surfaces) for the body assembly shall be incorporated into the testing equipment.

b. A calibrated mirror conforming to the dimensional and optical requirements of Drawing D10539515 shall be incorporated on the testing equipment. The mirror shall be positioned and adjusted to a 45 degree angle within 10 seconds so that a target image picked up at the mirror is projected 90 degrees within 10 seconds into the sighting system of the body assembly.

c. A target projector collimator having a clear aperture of at least 7.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned so that the optical axis of the collimator is in alinement with the line of sight emitting from the central aperture (geometric axis) of the body assembly. The target projector collimator shall have an appropriate calibrated target reticle pattern for projection into the optical system of the body assembly. The verticle reticle line of the target test reticle pattern shall be adjusted plumb. The target projector collimator test reticle shall be within the accuracies indicated for the following requirements:

Mirror repeatability	3 minutes of arc
Image tilt	"

The optical axis of the target projector collimator shall be alined parallel with respect to the optical axis of the viewing autocollimator and autocollimated to Datum Plane "Y" shown on Sheet 1 of Drawing F10539736, within 5 seconds. Unless otherwise specified the target image shall be adjusted for infinity focus.

d. A viewing autocollimator capable of measuring any line of sight deviations in the "X" and "Y" coordinates in minutes of arc shall be a part of the test equipment. The viewing autocollimator shall be positioned so that the optical axis of the viewing autocollimator is in alinement with the line of sight emitting from the central aperture (geometric axis) of the exit window of the unity power system and autocollimated to Datum Plane "A" within 5 seconds.

e. All set-up and calibration equipment with complete provisions necessary for certification and proof of accuracy of the test equipment shall be provided and shall become part of the testing equipment.

4.5.2.3.4 Orientation - Periscope: XM50.- The special testing equipment utilized to mount and orient the periscope for the accuracy tests shall conform to the following requirements:

a. The testing equipment shall be capable of mounting the periscope in its normal operating position, as depicted on Drawing F10539750.

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Adjustments shall be provided on the testing equipment that shall permit orientation of Datum Plane "A", shown on Sheet 1 of Drawing F10539736 in a horizontal plane within 10 seconds. The necessary mounting features (keys and locating surfaces) for the body assembly shall be incorporated into the testing equipment.

b. A target projector collimator having a clear aperture of at least 7.0 inches shall be a part of the test equipment. The target projector collimator shall be positioned so that the optical axis of the collimator is in alinement with the line of sight emitting from the central aperture (geometric axis) of the periscope head mirror. The target projector collimator shall have an appropriate calibrated target reticle pattern for projection into the optical system of the periscope. The verticle reticle line of the target test reticle pattern shall be adjusted plumb. The target projector collimator test reticle shall be within the accuracies indicated for the following requirements.

Elevation accuracy	5 seconds of arc
Plumb travel	"
Backlash	"
Excursion, elevation	10 seconds of arc
Excursion, depression	"
Mirror repeatability	3 minutes of arc
Image tilt	"
Parallelism of reticle and image	6 minutes of arc
Unity-power system	12 minutes of arc
Passive system	"

The optical axis of the target projector collimator shall be alined parallel with respect to the optical axis of the viewing autocollimator and autocollimated to Datum Plane "Y", shown on Sheet 1 of Drawing F10539736, within 5 seconds. Unless otherwise specified the target image shall be adjusted for infinity focus.

c. A viewing autocollimator capable of measuring any line of sight deviations in the "X" and "Y" coordinates in seconds of arc shall be a part of the test equipment. The viewing autocollimator shall be positioned so that the optical axis of the viewing autocollimator in alinement with the line of sight emitting from the central aperture (geometric axis) of the exit window of the unity power system and autocollimated to Datum Plane "A" within 5 seconds.

d. All set-up and calibration equipment with complete provisions, necessary for certification and proof of accuracy of the test equipment shall be provided and shall become part of the testing equipment.

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4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.7. Each item procured under this specification shall be exposed to the temperature extremes specified in 3.7 for a minimum of 5 hours at each extreme. Subsequent to the temperature exposures, the item shall be stabilized in an ambient temperature between +70° and 80° Fahrenheit and subjected to the appropriate visual, tactile, and audible examination of 4.6.3. The item shall conform to the requirements of 3.7. Caution shall be exercised during this test to avoid subjecting the item to thermal shock.

4.6.2 Extreme operating temperatures.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-19326 and the conditions of 3.8. Each item procured under this specification shall be exposed to the temperature extremes specified in 3.8 for a minimum of 5 hours at each extreme. At the end of each 5 hour period and while still at the temperature extreme, the item shall be subjected to the applicable tests specified in 4.6.4.3, 4.6.5.3, 4.6.5.4, 4.6.5.5, 4.6.5.6, 4.6.5.7, and 4.6.6.1.14. The item shall conform to the applicable requirement of 3.10.9, 3.11.4.1, 3.11.4.2, 3.11.4.3, 3.11.4.3.1, 3.11.4.3.2, and 3.12.7.3 at each temperature extreme specified in 3.8. Caution shall be exercised during this test to avoid subjecting the item to thermal shock.

4.6.3 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.9, shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. Each item procured under this specification shall be clamped on the vibrating equipment in its normal mounting or operating position by means of an appropriate adapter.

4.6.3.1 Head assembly.- The input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. The head assembly shall be mounted on the standard vibrating equipment specified. The vibrating motion shall be in the vertical plane for a period of 5 minutes ± 15 seconds. Subsequent to vibration, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials, to determine compliance with 3.9, prior to being subjected to the remaining tests specified herein. A visual and tactile examination shall be made for missing or damaged parts and to check functioning of moving parts for undue irregularities of movement, friction or looseness. An audible examination shall be made for loose internal optical, and mechanical parts. Any one of the possible defects defined above shall be cause to consider a head assembly defective, unless separate defects - per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Head assemblies which have passed the above examination shall be subjected to the tests of 4.6.4.1 to 4.6.4.5 inclusive to determine overall compliance with 3.9.

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4.6.3.2 Body assembly.- The body assembly shall be mounted on the standard vibrating equipment specified. The image tube shall be energized. The vibrating motion shall be in the vertical plane for a period of 5 minutes ± 15 seconds. During the vibration cycle, the image tube shall show no failure nor flicker of light emission when subjected to a visual examination. Subsequent to vibration, the body assembly shall be subjected to the visual, tactile and audible examination of 4.6.3.1 and shall show no evidence of physical failure, prior to being subjected to the remaining tests specified herein. Any one of the possible defects defined in 4.6.3.1 shall be cause to consider a body assembly defective, unless separate defects - per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Body assemblies which have passed the above examinations shall be subjected to the tests of 4.6.5.1.1 to 4.6.5.9 inclusive, to determine overall compliance with 3.9.

4.6.3.3 Reticle image projector.- The reticle image projector shall be mounted on the standard vibrating equipment specified. The reticle image projector lamp shall be energized. The vibrating motion shall be in the vertical plane for a period of 5 minutes ± 15 seconds. During the vibration cycle, the reticle projector lamp shall show no failure nor flicker of light emission when subjected to a visual examination. Subsequent to vibration, the reticle image projector shall be subjected to the visual, tactile and audible examination of 4.6.3.1 and shall show no evidence of physical failure, prior to being subjected to the remaining tests specified herein. Anyone of the possible defects defined in 4.6.3.1 shall be cause to consider a reticle image projector defective unless separate defects - per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Reticle image projectors which have passed the above examinations shall be subjected to the tests of 4.6.6.1.2 to 4.6.6.6 inclusive to determine overall compliance with 3.9.

4.6.3.4 Periscope: XM50.- The Periscope: XM50 shall be mounted on the standard vibrating equipment specified. The reticle image projector lamp and the image tube shall be energized. The vibrating motion shall be in the vertical plane for a period of 5 minutes ± 15 seconds. During the vibration cycle, the reticle image projector lamp and the image tube shall show no evidence of failure nor flicker of light emission when subjected to a visual examination. Subsequent to vibration, the periscope shall be subjected to the visual, tactile and audible examination of 4.6.3.1 and shall show no evidence of physical failure, prior to being subjected to the remaining tests specified herein. Any one of the possible defects defined in 4.6.3.1 shall be cause to consider a periscope defective unless separate defects - per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Periscopes which have passed the above examinations shall be subjected to the tests of 4.6.7.1.2 to 4.6.7.2.11 inclusive to determine overall compliance with 3.9.

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4.6.4 Head assembly.

4.6.4.1 Orientation.- The following tests shall be performed with the head assembly positioned on special testing equipment that conforms to the requirements of 4.5.2.3.2. The viewing autocollimator shall be autocollimated to Datum Plane "A", shown on Sheet 2 of Drawing F10539748. The optical axis of the target projector collimator shall be alined parallel with respect to the optical axis of the viewing autocollimator and autocollimated to a vertical plane containing the longitudinal axis of the 2 locating pins (MS9390-30) of the head assembly (see Sheet 1 of Drawing F10539749). Autocollimation and alinement shall be accomplished with the head assembly removed from the line of sight through the collimators.

4.6.4.1.1 Line of sight orientation.- Position the coupling to the angular value specified in 3.10.1. Direct a line of sight through the viewing autocollimator to the target. The intersecting point of the vertical and horizontal reticle lines of the viewing autocollimator shall be superimposed on the boresight mark of the projected target image within the tolerance specified in 3.10.1, as measured on the viewing autocollimator.

4.6.4.1.2 Excursion.- The testing equipment specified in 4.6.4.1 shall be utilized for this test except, the target projector collimator is not required. Auxiliary testing equipment employing an autocollimation device, affixed to the coupling of the head assembly shall be used in place of the target projector collimator. This testing equipment shall consist of an arm assembly with a calibrated optically flat front surface mirror assembly and an appropriate adapter that accommodates the coupling of the head assembly (see Figure 2). The mirror shall be autocollimated to the viewing autocollimator when the coupling is positioned 66° with respect to the mounting surface of the head assembly that mates to the body assembly. The reflecting (autocollimating) surface of the mirror shall lay in a plane parallel to a vertical plane containing the longitudinal axis of the 2 locating pins (MS9390-30) of the head assembly (see Sheet 1 of Drawing F10539749) and vertical centerline passing through the geometric axis of the coupling adapter. The reflecting (autocollimating) surface of the mirror shall also remain parallel to the vertical centerline of the coupling adapter during rotation through its full excursion of travel. Autocollimation of the mirror and the viewing autocollimator shall be accomplished with the head assembly removed from the line of sight. Position and secure the head assembly in the holding device with the auxiliary testing device affixed and oriented on the coupling of the head assembly. With backlash removed for direction of travel, position the coupling to the angular value specified in 3.10.1. Direct a line of sight through the viewing autocollimator and note the position of the boresight point (intersecting point of the vertical and horizontal reticle lines) of the real and reflected image. While observing the boresight point rotate the coupling, so that the line of sight is deflected 22° above to 18° below the position established in 4.6.3.1. Rotation of the coupling shall conform to the requirements specified in 3.10.2.

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4.6.4.1.3 Elevation accuracy.- Perform this test in conjunction with the test in 4.6.4.1.2 (Excursion). During rotation of the coupling the reflected reticle image shall not deviate from the real reticle image beyond the tolerance specified in 3.10.3, as measured on the viewing autocollimator. The head assembly shall then be subjected to the test specified in 4.6.4.1.4.

4.6.4.1.4 Plumb travel.- Perform this test in sequence to test in 4.6.4.1.3. Position the coupling to the angular value specified in 3.10.1. While observing the boresight point of the real and reflected image, rotate the coupling to the maximum elevation, then to the maximum depression. During this rotation, the reflected image shall not deviate in azimuth from the real image beyond the tolerance specified in 3.10.5. The head assembly shall then be subjected to the test specified in 4.6.4.1.5.

4.6.4.1.5 Backlash.- Perform this test in sequence to the test in 4.6.4.1.4. With backlash removed for direction of travel, position the coupling to the angular value specified in 3.10.1. Record the position of the boresight point of the reflected image as measured on the viewing autocollimator. Rotate the coupling approximately 10° in the same direction of travel, then reverse rotation of the coupling (with backlash included) until it is repositioned to the angular value specified in 3.10.1. Record this position of the boresight point of the reflected image, as measured on the viewing autocollimator. Any difference between the recorded reading is the amount of backlash. Backlash shall not exceed the limit specified in 3.10.4.

NOTE: During this operation (4.6.4.1.5), if overtravel occurs, the test shall be nullified and the procedure specified for backlash shall be performed again.

4.6.4.1.6 Luminous transmittance.- Conduct this test with the auxiliary testing equipment employing an autocollimation device removed from the coupling of the head assembly. A diffuse luminous source from a 2870°F tungsten lamp shall be positioned at the exit aperture. The transmittance measurement shall be made with an eye-corrected photometer. Compute the percentage of transmittance using the following:

$$\frac{\text{Luminance of source}}{\text{Luminance of source as seen through head}} \times 100\%$$

4.6.4.1.7 Transmittance at 0.85 micron - head assembly.- Conduct this test in sequence to the test in 4.6.4.1.6, utilizing the method specified therein. The transmission measurements shall be made with an image forming radiometer filtered for 0.85 micron. The filter of the radiometer shall conform to the requirements of 4.6.5.1.6. The transmittance shall be not less than the value specified in 3.10.8. Compute the percentage of transmittance using the following formula:

$$\frac{\text{Filter radiometer reading of target}}{\text{Filter radiometer reading as seen through head}} \times 100\%$$

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4.6.4.2 Resolution.- Position the head assembly on its side with the coupling immobilized at the angular value specified in 3.10.1. The test for resolution shall be performed utilizing a resolution target and a telescope of at least 80 power with an entrance pupil of at least 7.0 inches. The resolution target shall be placed on a wall target or in a collimator, positioned incident to the entrance window, and shall represent the angular subtense for the seconds of arc specified. Construction of the resolution target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. The target shall be illuminated by diffuse light from a 2870°K tungsten lamp. The resolution target shall have the detectable line structure and proper line count when observed through the telescope positioned at the exit aperture of the head assembly. Resolution shall meet the requirements of 3.10.6.

4.6.4.3 Operability.- Standard torque measuring equipment capable of measuring the torque values specified in 3.10.9 shall be utilized for this test. An appropriate adapter, to accommodate the input coupling shall be used with the torque measuring equipment. Torque shall be measured while rotating the input coupling through its full range of travel. The running torque required to rotate the input coupling shall conform to the applicable requirements of 3.10.9 for the temperatures indicated.

4.6.4.4 Cleanliness.- The optical surfaces of the head assembly shall be inspected for cleanliness in accordance MIL-O-13830. The head assembly shall conform to the cleanliness requirements of 3.10.10.

4.6.4.5 Sealing.- Prior to the internal pressurizing of the head assembly, a cover capable of sealing the bottom opening of the head assembly, shall be secured to the head mounting surface. The internal pressurizing of the head assembly to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.1. The pressure shall be applied to the head assembly through the valve provided in the upper housing. Pressurize the head assembly with dry nitrogen gas, having a dewpoint at least as low as minus 25 degrees Fahrenheit, in accordance with the procedures for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the head assembly for a period of 5 minutes ± 15 seconds. During this time period, the input coupling shall be rotated through one full excursion of its articulation range. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment to determine compliance with 3.10.11.

4.6.5 Body assembly.

4.6.5.1 Orientation, optical characteristics.- The following tests shall be performed with the body assembly positioned on a holding device that permits the body assembly to be mounted as required in 3.11 and pivoted in the horizontal plane. A calibrated mirror conforming to the dimensional and optical requirements of Drawing D10539515, shall be incorporated on the holding device. The mirror shall be positioned at a 45 degree angle so that a target image picked up at the mirror is turned 90 degrees and projected into the sighting system of the body assembly.

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Unless otherwise specified the holding device shall be immobilized in a zero aximuth position, the image tube shall be removed from the body assembly, and the clear filter shall be positioned in the optical train.

NOTE: The test values for all characteristics shall be recorded and made a part of the inspection record.

4.6.5.1.1 Contrast transfer - on-axis, objective system.- Perform this test using a 2870°K light source, diffuse screen, frequency pattern (see Figure 3), collimator, an S-20 extended red photo-detector with slit (see Figure 4), and an "X-Y" recorder set up and oriented as in Figure 5. Position the photo-detector in the tube cavity of the body assembly utilizing an appropriate adapter. Focus the image of the frequency pattern at the slit. Adjust the "X-Y" recorder for zero level by blocking the entrance of all light to the photo-detector. Measurements shall be made at 5 frequencies in the image plane. Position the photo-detector on axis and advance the frequency pattern, in a horizontal direction, through the five frequencies at a constant rate such that no signal damping due to response time of the photo-detector occurs, and obtain recordings similar to that shown in Figure 6. Calculate the on-axis contrast transfer to vertical lines using the formula $\frac{W - B}{W + B}$ from measured data. Record the contrast transfer values for the vertical lines. Rotate the frequency pattern and slit 90%. Measure, calculate, and record the on-axis contrast transfer values for the horizontal lines. Contrast transfer on-axis shall conform to the applicable requirements specified in 3.11.3.1.1.

NOTE: Special care must be taken before test data is made to avoid possible introduction of contrast transfer losses resulting from the test instrument.

4.6.5.1.2 Contrast transfer - off-axis, objective system.- Perform this test in sequence to the test in 4.6.5.1.1 utilizing the testing equipment and test method specified therein. Off-axis measurements shall be accomplished by translating the microscope objective and slit (Figure 4) perpendicular to the optical axis without refocusing. Measure, calculate, and record the off-axis contrast transfer values for the vertical lines with the microscope objective and slit translate off-axis to the left from on-axis. Translate the microscope objective and slit downward from on-axis and repeat the measurements, calculations, and recording for both vertical lines and horizontal lines. Contrast transfer off-axis values shall conform to the applicable requirements specified in 3.11.3.1.1 (see 4.6.5.1.1 note).

4.6.5.1.3 Contrast transfer - on - and off-axis, eyepiece and erecting system.- Perform this test using a 2870°K light source, diffuse screen, frequency pattern (see Figure 3), collimator, an eye corrected photo-detector with slit (see Figure 7), and an "X-Y" recorder set up and oriented as in Figure 8. An appropriate adapter shall be utilized to position the photo-detector in the tube cavity of the periscope. Set the eyepiece of the body assembly to zero diopters. Perform this test using the test methods and procedures specified in 4.6.5.1.1 and 4.6.5.1.2. Measure, calculate, and record the on-axis contrast transfer values for

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both vertical lines and horizontal lines. Measure, calculate, and record the off-axis contrast transfer values for the vertical lines with the microscope objective and slit translated off-axis to the right from on-axis. Translate the microscope objective and lit upward and repeat the measurements, calculations, and recording for both vertical lines and horizontal lines. Contrast transfer on-axis and off-axis shall conform to the applicable requirements specified in 3.11.3.1.2 (see 4.6.5.1.1 note).

4.6.5.1.4 Radiant transmittance, objective system - without filters.- A diffuse surface with a luminance of not less than 100 foot-lamberts shall be positioned at the image plane of the objective system (image tube cavity) of the body assembly. The light source shall be from a 2870°K tungsten lamp (see Figure 9). The entire entrance pupil of the head assembly shall be scanned with an eye corrected photometer. The luminous transmittance of the image, measured at the entrance pupil at any point, shall be not less than the value specified in 3.11.3.2.1a. Percentage of transmission shall be defined as:

$$\frac{\text{Apparant luminance as seen thru body assembly}}{\text{Luminance of target}} \times 100\%$$

4.6.5.1.5 Radiant transmittance, objective system - with filters.- Perform this test in sequence to the test in 4.6.5.1.4 with each specified filter positioned in the optical train, in turn. The ratio of luminous transmittance of the image for the filter positions specified, as measured at the entrance pupil, shall be within the limits of the values specified in 3.11.3.2.1b.

4.6.5.1.6 Transmittance at 0.85 micron, objective system - without filters.- Perform this test in sequence to the test in 4.6.5.1.5. Transmittance at 0.85 micron shall be not less than the value specified in 3.11.3.2.2a as measured with an image forming radiometer filtered for 0.85 micron. The filter utilized with the radiometer shall have a peak transmission of not less than 30% at 0.85 micron, and shall have a bandwidth of ± 0.01 micron at 1/3 of peak.

4.6.5.1.7 Transmittance at 0.85 micron, objective system - with filters.- Perform this test in sequence to the test in 4.6.5.1.6 with each specified filter positioned in the optical train, in turn. The ratio of radiant transmittance of the image for the filter position specified, as measured at the entrance pupil, shall be within the limits of the value specified in 3.11.3.2.2b.

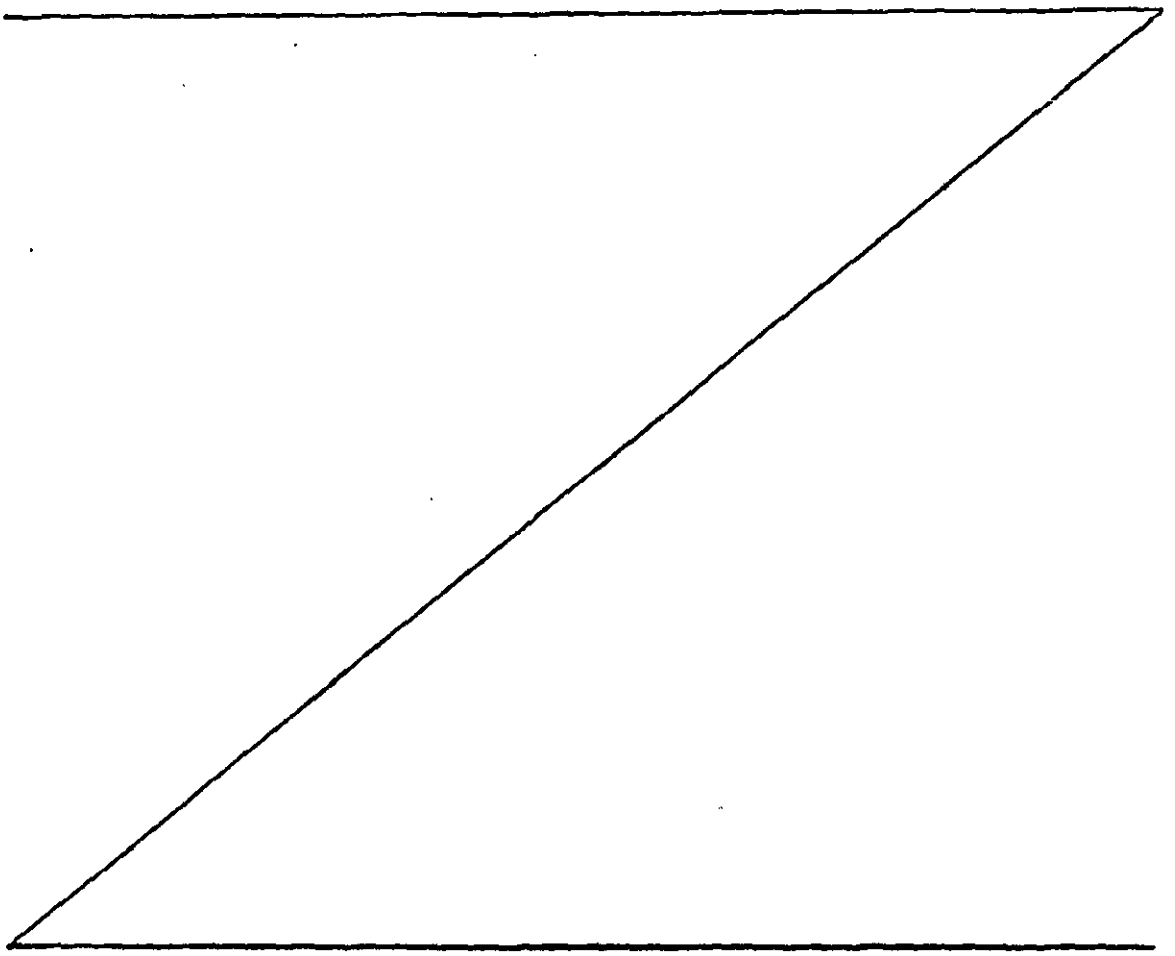
4.6.5.1.8 Eyepiece and erecting system, luminous transmittance.- A diffuse surface with a luminance of not less than 100 foot-lamberts shall be positioned in the object plane (image tube cavity) of the eye telescope. The luminance of the target as seen at the eyepiece of the body assembly shall be not less than the value specified in 3.11.3.3 when measured with an eye-corrected photometer.

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4.6.5.1.9 Resolution on-axis, objective system.- Position a resolution pattern on a reticle placed in the focal plane of the objective system of the body assembly (image tube cavity). Construction of the resolution pattern reticle shall be as shown in Figure 10. The resolution pattern shall have the detectable line structure and proper line count when observed through a telescope positioned at, and normal to, the entrance aperture of the body assembly (see Figure 11). The telescope shall be of at least 90 power with an entrance pupil of at least 7.0 inches. Resolution on-axis shall meet the requirements of 3.11.3.4.1a.

4.6.5.1.10 Resolution off-axis, objective system.- Perform this test as specified in 4.6.5.1.9, except that the viewing telescope shall be rotated off-axis the required distance. Resolution off-axis shall meet the requirements of 3.11.3.4.1b.

4.6.5.1.11 Objective T-number - on-axis.- The effective T-number will be determined by flooding the image plane of the body assembly objective with a known luminosity and measuring the illumination of the projected image on a target placed in front of the body assembly. A



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luminous source providing a diffuse luminosity of approximately 100 foot-lamberts over a circular area 1.0 inch in diameter shall be positioned in the image tube cavity by means of a suitable adapter. A reticle shall be mounted between the luminous source and the objective lens and positioned to provide a sharply focused image at a distance of 10.0 feet from the entrance window of the body assembly. The reticle pattern shall be as indicated in Figure 12. Luminance measurements shall be made from a magnesium oxide target placed within the center cross mark of the reticle image at 10.0 feet. The objective T-number shall not exceed the value specified in 3.11.3.4.1.1a.

4.6.5.1.12 Objective T-number - off-axis.- Perform this test as in 4.6.5.1.11. The objective T-number shall be measured at the 4 peripheral cross marks of the reticle image. The objective T-numbers shall not exceed the value specified in 3.11.3.4.1.1b.

4.6.5.1.13 Resolution - unity-power system.- The test for resolution shall be performed utilizing an auxiliary telescope conforming to the requirements of 3.11.1.1, and a resolving power chart on a wall target or positioned in a collimator. The resolving power target shall represent the angular subtense for the minutes of arc specified. Construction of the target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. The target shall be illuminated by diffuse light from a 2870°K tungsten lamp. The target shall have the detectable line structure and proper line count when observed through the telescope positioned at the exit aperture of the unity power system. Resolution shall meet the requirements of 3.11.1.1.

4.6.5.1.14 Resolution - daylight system.- Perform this test in sequence with 4.6.5.1.13, utilizing an auxiliary telescope conforming to the requirements of 3.11.2.1 and a resolving power chart placed on a wall target or positioned in a collimator. The resolving power target shall represent the angular subtense for the seconds of arc specified in 3.11.2.1. Construction and illumination of the resolving power chart shall be as specified in 4.6.5.1.13. The auxiliary telescope shall be positioned at the eyepiece of the body assembly. The focus of the auxiliary telescope shall be established at one best focus setting for simultaneous viewing of the detectable line structure and proper line count of the resolution target. Resolution shall meet the requirements of 3.11.2.1 when observed at one focus, through the auxiliary telescope.

4.6.5.1.15 Body assembly with image tube installed.- Unless otherwise specified, the following tests shall be performed with the image tube energized. The voltage required to energize the image tube shall be supplied from a power source capable of supplying and maintaining the required voltage. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting these tests.

NOTE: Government furnished image intensifier tubes.- When the image intensifier tube is provided by the Government each tube shall be subjected to a verification reinspection by the contractor for the following

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characteristics, prior to insertion into the body assembly:

(a) Luminance gain (+69.8° to 80.6°F)
(b) Image alinement
(c) Center resolution
(d) Peripheral resolution
(e) Breakdown
(f) Equivalent background input (illumination)
(g) Contrast transfer

The characteristics are defined in the Image Intensifier Tube Purchase Description referenced in 3.11.3.5. Examinations and tests shall be in accordance with the Quality Assurance Provisions of the referenced purchase description.

4.6.5.1.16 Resolution - passive system, on-axis.- Perform this test using a projected or mechanical target. The effective range shall be not less than 50 meters. The radiance presented to the periscope shall have a color temperature of 2870°K. The background luminance and contrast (high and low) of the target shall be as specified in 3.11.3.5. Construction of the target shall conform to the USAF 1951 Resolution Pattern in accordance with MIL-STD-150. Adjust the focus of the periscope. The resolution target shall have the detectable line structure and proper line count when observed through the eyepiece with the unaided eye. Resolution on-axis for high and low contrast shall meet the requirements specified in 3.11.3.5. Test data results from this test are CONFIDENTIAL.

4.6.5.1.17 Resolution - passive system, on-axis.- Rotate the periscope off-axis 2.0 degrees and repeat the test specified in 4.6.5.1.16. Resolution off-axis for high contrast shall meet the requirements specified in 3.11.3.5. Test data results from this test are CONFIDENTIAL.

4.6.5.1.18 Contrast transfer - on-axis.- Perform this test using a 2870°K tungsten light source, diffuse screen, frequency pattern (see Figure 3), collimator, eye-corrected photo-detector with slit, and an "X-Y" recorder set up and oriented as in Figure 13. Illuminate the frequency pattern by means of a 2870°K tungsten lamp and focus the image of the frequency pattern at the slit. Adjust the "X-Y" recorder for zero level and perform the test, as in 4.6.5.1.1, by advancing the frequency pattern through the three frequencies horizontally and then vertically as in 4.6.5.1.1. Calculate the contrast transfer using the formula $\frac{W-B}{W+B}$ from measured data. Compute the required contrast transfer using the applicable values in 3.11.3.6 and the values of U_T. Contrast transfer on-axis shall conform to the applicable requirements of 3.11.3.6. Special care must be taken before test data is made to avoid the possible introduction of contrast transfer losses resulting from the test instrument. Test data results from this test are CONFIDENTIAL.

4.6.5.1.19 Contrast transfer off-axis.- Perform this test in sequence to the test in 4.6.5.1.18 utilizing the test equipment and test method specified therein. Off-axis measurements, left and right, shall be accomplished by moving the photo-detector to off-axis points by translation perpendicular to the optical axis without refocusing. Measure and record the 3 frequencies off-axis left, and the 3 frequencies off-axis, right, contrast transfer measurements for the horizontal plane. Measure

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and record the 3 frequencies off-axis left, and 3 frequencies off-axis, right, contrast measurements for the vertical plane. Contrast transfer off-axis shall conform to the applicable requirements specified in 3.11.3.6. (TEST DATA RESULTS FROM THIS TEST ARE CONFIDENTIAL).

4.6.5.1.20 Luminance gain.- Perform this test using a 2870°K tungsten light souree, a diffuser with mask, collimator, an eye-corrected photo-detector with a 7.0 to 9.5 mm entrance pupil, and a calibrated foot-lambert meter set up and oriented as in Figure 14. The calibrated foot-lambert meter shall be used to measure luminance gain by comparing the luminance seen by the periscope with the luminance at the periscope eyepiece. The light-source collimator shall be calibrated to present to the periscope a uniform target luminance of 1.5×10^{-3} foot-lamberts, with illumination from a 2870°K tungsten lamp, subtending a solid angle of 4.5 degrees of periscope field of view centered on the optical axis. Energize the periscope and measure the luminance at the eyepiece with an eye-corrected brightness meter which has been calibrated for light from 2870°K tungsten lamp and has an entrance pupil of between 7.0 mm and 9.5 mm. The gain is the ratio of the luminance at the eyepiece divided by the target luminance. The luminance gain shall be not less than the value specified in 3.11.3.7. (TEST DATA RESULTS FROM THIS TEST ARE CONFIDENTIAL).

4.6.5.1.21 Stray light luminance.- Perform this test using a calibrated foot-lambert meter, an eye-corrected photo-detector, a 2870°K tungsten light source, a surface of white surround conforming to 3.11.3.8, and a disk with a black surface luminance of less than 10^{-6} foot-lamberts set up and oriented as in Figure 15. The instrument described above and in Figure 13 shall be used to measure the stray light luminance. The 2870°K tungsten light source shall be so positioned as to provide a nearly uniform luminance of 5 times 10^{-3} foot-lamberts over the surface of the white surround. The eye-corrected photo-detector shall have an aperture of 7.0 mm to 9.5 mm. Adjust the body assembly for infinity focus. Compare the eyepiece luminance with light source off and light source on. Any change in eyepiece luminance, in the measurements with the light source off and the light source on, shall not exceed the tolerance specified in 3.11.3.8.

4.6.5.1.22 Field of view.- Project a cross-line test pattern onto the objective lens of the periscope by means of a diffuse 2870°K tungsten lamp. The body assembly shall then be pivoted either to the left or right until the center point of the test pattern seen through the body assembly is at the edge of the field of view. The body assembly shall then be pivoted in the opposite direction until the center point of the cross is at the opposite edge of the field of view. The total angle through which the body assembly was pivoted shall be measured. The field of view of the body assembly shall be not less than the value specified in 3.11.3.9. Subsequent to this test, return the body assembly to zero azimuth position and immobilize the holding device.

4.6.5.1.23 Definition and visibility, projected reticle pattern, clear filter.- Perform this test with diffuse illuminance from a 2870°K tungsten lamp presented to the entrance window of the body assembly. The luminance shall not exceed the value specified in 3.11.3.10.1. A reticle image projector collimator having a clear aperture of at least 2.0 inches

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shall be utilized with the holding device, to project an appropriate calibrated reticle pattern incident to the Window, Drawing B10539305, of the body assembly. The vertical reticle line of the collimator shall be adjusted plumb. The image of the projected reticle pattern shall conform to the requirements of 3.11.3.10.1.

4.6.5.1.24 Definition and visibility of projected reticle pattern - clear, medium and dark filters.- Perform this test with the conditions required in 3.11.3.10.1 and 3.11.3.10.2 established. The image of the reticle pattern shall conform to the requirements of 3.11.3.10.1 and 3.11.3.10.2 when each specified filter is positioned in the optical train in sequence.

4.6.5.1.25 Flicker - image steadiness.- Perform this test with a diffuse illumination from a 2870°K tungsten lamp presented to the entrance window of the body assembly. Project the reticle pattern into the optical train. Position an auxiliary telescope at the eyepiece of the body assembly. The auxiliary telescope shall contain a cross-line reticle and be capable of focusing on the screen of the image tube. Observe through the auxiliary telescope and superimpose the horizontal and vertical reticle lines on the reticle pattern lines on the screen of the image tube. Observe the target image for a time period of not less than one minute. During this one minute time period there shall be no visivle flicker or movement of the image as specified in 3.11.3.11.1. Image movement shall be determined by means of the target image with respect to the auxiliary telescope reticle pattern.

4.6.5.1.26 Sparking - image steadiness.- Perform this test subsequent to, and utilizing the procedure, specified in 4.6.5.1.25. During the one minute period of observation there shall be no visible or audible evidence of sparking as specified in 3.11.3.11.2.

4.6.5.1.27 Cleanliness - passive system.- Perform this test with diffuse illumination from a 2870°K tungsten lamp presented to the entrance window of the body assembly. Project the reticle pattern into the optical train. Visually inspect the nine-power system through the body assembly eyepiece. The nine-power system shall conform to the requirements of 3.11.5.2.

4.6.5.1.28 Cleanliness - unity-power system.- Perform this test with the body assembly switch in the OFF position. Position a diffuse light source at the entrance window of the body assembly. Visually inspect the unity-power optical system through the exit window of the system. The unity-power system shall conform to the requirements of 3.11.5.1.

4.6.5.1.29 Cleanliness - daylight system.- Perform this test in sequence to 4.6.5.1.28 with the body assembly switch in the OFF position and utilizing the diffuse light source as specified therein. Visually inspect the daylight optical system through the eyepiece of the body assembly. The daylight optical system shall conform to the requirement of 3.11.5.3.

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4.6.5.2 Orientation - optical accuracies.- The following tests shall be conducted with the body assembly positioned as specified in 3.11 on special testing equipment conforming to 4.5.2.3.3. The viewing autocollimator shall be autocollimated to Datum Plane "A" shown on Sheet 1 of Drawing F10539736. The optical axis of the target projector collimator shall be alined parallel with respect to the optical axis of the viewing autocollimator and autocollimated to Datum Plane "Y" (see Sheet 1 of Drawing F10539736). The reticle pattern of the target collimator shall be projected by means of diffuse luminance from a 2870°K tungsten lamp. Luminance shall not exceed 1×10^{-1} foot-lamberts. Autocollimation and alinement shall be accomplished with the body assembly removed from the line of sight through the collimators. Unless otherwise specified the image tube shall not be energized.

4.6.5.2.1 Mirror repeatability - unity power system.- Position the mirror in the optical train by means of the upper introduction lever. Direct a line of sight through the viewing autocollimator to the target, and record the position of the intersecting point of the vertical and horizontal reticle lines of the viewing autocollimator with respect to the intersecting point of vertical and horizontal reticle lines of the target. Remove the mirror out of the optical train by means of the upper introduction lever. Again, position the mirror in the optical train by means of the upper introduction lever. Direct a line of sight through the viewing autocollimator and record the position of the intersecting point of the vertical and horizontal reticle lines of the viewing autocollimator with respect to the intersecting point of the vertical and horizontal reticle lines of the target. Any difference between the recorded readings shall conform to the requirements of 3.11.1.2. Repeat this test a minimum of three times. The upper introduction lever shall be locked in detent at each position (in and out).

4.6.5.2.2 Image tilt - daylight system.- Position an auxiliary telescope of at least 3 power (3X) containing a cross-line reticle pattern at the eyepiece of the body assembly. Plumb the vertical reticle line of the telescope and superimpose the reticle line on the vertical target line image. The image of the vertical target line shall be parallel to the telescope vertical reticle line within the tolerance specified in 3.11.2.2 as measured on the target.

4.6.5.2.3 Eyepiece focus - daylight system.- A calibrated dioptometer shall be utilized for measuring the accuracy of zero diopter setting of the diopter knob. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the diopter knob of the body assembly to indicate zero diopter. Position the dioptometer at the eyepiece of the body assembly. Observe through the dioptometer, obtain best focus of the projected target image by adjustment of the body assembly diopter scale. The target image shall be in focus within the tolerance specified in 3.11.2.3, as measured on the diopter scale of the body assembly. Position the body assembly diopter scale to zero diopter as indicated on the scale with reference to the index mark. Rotate the diopter scale to its clockwise stop, note the diopter setting on the scale, then rotate the diopter scale to its counterclockwise stop, note the diopter setting on the scale. The diopter movement shall conform to the requirements of 3.11.2.3.

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4.6.5.2.4 Eyepiece backlash - daylight system.- With backlash excluded for direction of travel, rotate the diopter scale in a clockwise direction and obtain best focus of the infinity target projector image. Record the diopter scale reading. Continue rotation of the diopter scale to its stop, then with backlash included, rotate the scale counterclockwise and obtain best focus of the target image without overtravel. Record the diopter scale reading. The difference between the first and second recorded scale readings is the amount of backlash in the diopter movement. Backlash in the diopter movement shall not exceed the tolerance specified in 3.11.2.4.

NOTE: Should overtravel occur during rotation of the diopter scale the test is nullified. When overtravel occurs, the test for backlash shall be repeated.

4.6.5.2.5 Objective focus.- This test shall be performed with the image tube energized. The voltage required to energize the image tube shall be supplied from a power source capable of supplying and maintaining the required voltage. The test for short range focus requirements of the body assembly shall be performed with the target image adjusted to a focus of 50 meters. With the diopter knob positioned for best focus of the phosphor grain on the screen of the image tube, rotate the focusing knob of the body assembly to the short range setting. The body assembly shall conform to the requirements specified in 3.11.3.12. The test for infinity (long range) requirements of the periscope shall be performed with the target image adjusted for infinity focus. Rotate the focusing knob to the infinity setting, the periscope shall conform to the requirements specified in 3.11.3.12. The periscope shall then be subjected to the test specified in 4.6.5.2.6.

4.6.5.2.6 Diopter change.- Adjust the periscope diopter knob for best focus of the phosphor grain on the screen of the image tube. Record the diopter scale reading. Rotate the focusing knob through the range specified in 3.11.3.2. Any adjustment of the diopter knob to obtain best focus of the target image, due to rotation of the focusing knob, shall not exceed the tolerance specified in 3.11.3.12.1 as measured on the diopter scale.

4.6.5.3 Operability - diopter movement.- Standard torque measuring equipment capable of measuring the torque values specified in 3.11.4.1 shall be utilized for this test. An appropriate adapter, to accommodate the diopter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the diopter knob through its full range of travel. The running torque required to rotate the diopter knob shall conform to the applicable requirements of 3.11.4.1 for the temperatures indicated.

4.6.5.4 Operability - passive, daylight and unity power introduction levers.- Standard force measuring equipment capable of measuring the force values specified in 3.11.4.2 shall be utilized for this test. An appropriate adapter, to accommodate the detent knob of the introduction levers shall be used with the force measuring equipment. Force shall be measured

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while actuating the detent knob through its full range of travel. The force required to disengage the detents from each lock position (in and out) shall conform to the applicable requirements of 3.11.4.2 for temperatures indicated. The introduction levers shall be subjected to a tactile examination and shall conform to the applicable requirements of 3.11.4.2 for the temperatures indicated. The filter knob shall be subjected to a tactile examination to determine over all compliance with 3.11.4.2.

4.6.5.5 Operability-micro-switch.- Standard torque measuring equipment capable of measuring the torque values specified in 3.11.4.3.2 shall be utilized for this test. An appropriate adapter, to accommodate the filter knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the filter knob through its entire excursion. The running torque required to rotate the filter knob shall conform to the applicable requirements of 3.11.4.3.2 for the temperatures indicated. While still at the temperature exposures, a continuity test of the micro-switch circuitry shall be made using standard measuring equipment. Continuity shall be checked through the entire excursion of the filter knob, while observing the standard measuring equipment to determine overall compliance with 3.11.4.3.

4.6.5.6 Operability - rheostat control.- Standard torque measuring equipment capable of measuring the torque values specified in 3.11.4.3.1 shall be utilized for this test. An appropriate adapter, to accommodate the rheostat control knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the rheostat control knob through its entire excursion. The running torque required to rotate the rheostat control knob shall conform to the applicable requirements of 3.11.4.3.1 for the temperatures indicated.

4.6.5.7 Operability - objective focus control.- Standard torque measuring equipment capable of measuring the torque values specified in 3.11.4.3.2 shall be utilized for this test. An appropriate adapter, to accommodate the objective focus control knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the objective focus control knob through its entire excursion. The running torque required to rotate the objective focus control knob shall conform to the applicable requirements of 3.11.4.3.2 for the temperatures indicated.

4.6.5.8 Interior atmosphere.- A dew point tester shall be used to test the dew point of the interior atmosphere of the body assembly. Upon satisfactory completion of the sealing test, the internal pressure of the body assembly shall be reduced to 1 psig. The body assembly shall then be flushed with dry nitrogen gas. The flow of gas into the body assembly shall be controlled by a regulator that will prevent the internal pressure in the body assembly from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the body assembly has a dew point at least as low as minus 25°F, as measured with the dew point tester. The body assembly shall then be sealed. The internal atmosphere of the body assembly shall conform to the requirements of 3.11.7.

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4.6.5.9 Sealing.- The internal pressurizing of the body assembly to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.1. Pressurize the body assembly with dry nitrogen gas, to the value specified in 3.11.6 in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the body assembly for a period of 5 minutes ± 15 seconds. During this time period, the introduction levers and the eyepiece, filter, rheostat, and focus control knobs shall be rotated to their clockwise and counterclockwise stops. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment, to determine compliance with 3.11.6.

4.6.6 Reticle image projector.

4.6.6.1 Orientation.- Special testing equipment, capable of positioning the reticle image projector as specified in 3.12.1, shall be utilized for the following tests. Unless otherwise specified the testing equipment shall also be provided with an observation telescope positioned at the exit window of the reticle projector. The telescope shall be at least 10 power (10X) with an entrance pupil of at least 2.5 inches and shall have a field of view of at least 70 mils. The reticle of the telescope shall have an appropriate calibrated reticle pattern for measuring the optical accuracies of the reticle image projector. The reticle shall contain a horizontal and vertical line with graduation marks at 0.5 mil increments on the lines for 35 mils on each side of the horizontal and vertical reticle cross lines. The telescope shall be adjusted for zero parallax at infinity focus. The vertical reticle line shall be adjusted plumb.

4.6.6.1.1 Electrical-optical accuracy.- Unless otherwise specified, to comply with 3.12.2 the projector shall be energized. For sources and measurements, the test equipment referenced in 4.6.6.1.13 shall apply.

4.6.6.1.2 Horizontal excursion.- The test shall be performed with the projector mounted on a testing device as depicted in 4.6.6.1 and in conjunction with the conditions specified in 4.6.6.1.1. With the boresight cross of the reticle projector positioned as specified in 3.12.2.1 and through movement of the reticle, the boresight cross shall move as specified in 3.12.2.1 as observed and measured on the observation telescope.

4.6.6.1.3 Vertical excursion.- With the boresight cross of the reticle projector positioned as specified in 3.12.2.2 and through movement of the reticle, the boresight cross shall travel as specified in 3.12.2.2 as observed and measured on the observation telescope.

4.6.6.1.4 Horizontal reticle travel.- This test may be performed in conjunction with 4.6.6.1.2 using the appropriately marked reticle line for the 28 mil horizontal excursion. Position the horizontal reticle boresight cross line spread coincident to the horizontal target line. Through movement of the reticle, the boresight cross shall not deviate from the horizontal target line in excess of the tolerance specified in 3.12.2.3 as measured on the observation telescope.

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4.6.6.1.5 Vertical reticle travel.- The test may be performed in conjunction with 4.6.6.1.3 using the appropriately marked target for the 26.67 mil and 55 mil vertical excursion. Position the projector reticle boresight point coincident to the intersecting point of the vertical and horizontal reticle target image lines. Through movement of the reticle, the boresight point shall not deviate from the vertical target line image in excess of the tolerance specified in 3.12.2.4 as measured on the observation telescope.

4.6.6.1.6 Lateral displacement.- Establish coincidence of the horizontal line of the boresight cross of the projector reticle with the horizontal target line image. Move the projector reticle through its full excursion in each direction. During the movement of the reticle, the boresight mark shall not deviate in a direction perpendicular to the intended movement in excess of the tolerance specified in 3.12.2.5.

4.6.6.1.7 Backlash.- With backlash excluded for direction of travel establish coincidence of the reticle boresight point with the boresight mark of the target image. Record this position as measured on the observation telescope. Move the reticle to its full excursion in the same direction, then with backlash included, reestablish coincidence of the boresight point and boresight mark without overtravel. Record the position of the boresight point with respect to the target image. Any difference between the two recorded readings is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.12.2.6, as measured on the observation telescope. Should overtravel occur during the movement of the projector reticle, the test is nullified. When overtravel occurs, the test for backlash shall be repeated.

4.6.6.1.8 Stability.- This test shall be performed in sequence to the test in 4.6.6.1.7. The boresight cross of the reticle projector shall be positioned to any known point within the excursion specified in 3.12.2.1 and 3.12.2.2. The reticle projector shall meet the requirements specified in 3.12.2.7 as observed and measured on the observation telescope.

4.6.6.1.9 Reticle focus.- The testing equipment specified in 4.6.6.1 shall be utilized for this test with the exception of the observation telescope. A calibrated dioptometer with a magnification of at least three power (3X) shall be used in lieu of the observation telescope. Set the dioptometer's objective diopter scale to zero, and adjust the dioptometer's eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the exit window of the reticle projector. Observe through the dioptometer and obtain best focus of the image projector reticle, by adjustment of the dioptometer's objective diopter scale. The image projector reticle shall be in sharp focus within the tolerance specified in 3.12.3 as measured on the dioptometer's objective diopter scale.

4.6.6.1.10 Reticle accuracy.- The angular subtense between the boresight cross and the indicated maximum graduations of the image projector reticle pattern shall not exceed the tolerance specified in 3.12.4 as measured on the observation telescope.

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4.6.6.1.11 Parallax.- Direct a line of sight through the observation telescope reticle to the image projector reticle. Parallax is any apparent displacement of the complete reticle pattern of the image projector reticle in relation to the reticle pattern of the observation telescope when the observer's head is moved from side to side or up and down. Parallax shall not exceed the tolerance specified in 3.12.5.

4.6.6.1.12 Parallax of reticle and image.- Direct a line of sight through the observation telescope and obtain coincidence of one extremity of the image projector's horizontal reticle line with that of the observation telescope reticle. Then observe for coincidence at the opposite extremity of the image projector's horizontal reticle line. The total spread of the image projector's horizontal reticle line through the boresight cross shall be parallel to the horizontal line of the observation telescope within the tolerance specified in 3.12.6.

4.6.6.1.13 Electrical phasing.- To conform with 3.12.7, the projector shall be connected to sources capable of supplying the voltages depicted on Drawing F10540243 in conjunction with an oscilloscope, Tektronix Model 28B or equal and a ratio standard Gertsch AC-DC Ratio Standard Model 1001 or equal, for observing null reading, and capable of maintaining the in-phase and out-of-phase relationships. The applied potential shall have an alternating current (ac) frequency of 400 cycles, plus or minus 5 percent.

4.6.6.1.13.1 Horizontal movement.- When the voltages are applied as specified in 3.12.7.1, check the phase relationship between pins 5 and 2 of J1 and observe that they are in phase. The movement of the reticle shall be to the right to conform to the direction specified in 3.12.7.1. A 180 degree reversal of the phase relationship between pins 5 and 2 shall cause the reticle to move in the opposite direction to conform with the requirement specified in 3.12.7.1.

4.6.6.1.13.2 Vertical movement.- When the voltages are applied as specified in 3.12.7.2, check the phase relationship between pins 9 and 2 of J1 and observe that they are in phase. The movement of the reticle shall be in depression to conform to the direction specified in 3.12.7.2. A 180 degree reversal of the phase relationship between pins 9 and 2 shall cause the reticle to move in the opposite direction to conform with the requirement specified in 3.12.7.2.

4.6.6.1.13.3 Slewing speed.- A standard type time measuring device shall be utilized for this test. The time shall be checked from the instant of voltage application to the maximum extent of travel of the reticle in both directions in conformance with the requirements of 3.12.7.3, and shall not exceed the time limit at the temperatures specified in 3.12.7.3.

4.6.6.1.14 Illumination.- This test shall be performed in a darkened area. Apply a 24 volt dc 400 cycles to pins 1b and 6 (ground) of connection J1. Observe the image projector reticle. All reticle lines and graduations of the image projector pattern shall be clearly defined while the reticle lamp is illuminated from minimum to maximum intensity by means of the rheostat control knob.

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4.6.6.2 Dialectric strength.- Perform this test in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment to determine conformance with the requirements of 3.12.9. Lamps shall be removed for this test.

4.6.6.3 Insulation resistance.- Immediately after satisfactorily withstanding the dielectric strength test, the projector shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B of MIL-STD-202. Measurements between the insulating points and ground shall be made utilizing standard measuring equipment to determine conformance with the requirements of 3.2.10. Lamps shall be removed for this test.

4.6.6.4 Cleanliness.- The optical system of the projector shall be examined through the window end, with the unaided eye. There shall be no evidence of dirt to obscure any portion of the projected markings of the reticle pattern when the reticle lamp is energized from minimum to maximum intensity.

4.6.6.5 Sealing.- The internal pressurizing of the reticle image projector to test sealing, shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied to the projector through the valve provided in the housing. Pressurize the projector with dry nitrogen gas, having a dew point of at least as low as minus 25 degrees Fahrenheit, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the projector for a period of 5 minutes plus or minus 15 seconds. At the end of the 5 minutes period there shall be no evidence of leakage, as indicated on the pressure gage of the testing equipment to determine compliance with 3.12.12.

4.6.6.6 Interior atmosphere.- A dew point tester shall be used to test the dew point of the interior atmosphere of the projector. Upon satisfactory completion of the sealing test, the internal pressure of the projector shall be reduced to 1 psig. The projector shall then be flushed with dry nitrogen gas. The flow of gas into the projector shall be controlled by a regulator that will prevent the internal pressure in the projector from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the reticle projector has a dew point at least as low as minus 25°F, as measured with the dew point tester. The projector shall then be sealed and the internal pressure of the projector shall be reduced to a pressure between 0.10 and 0.25 psig. The external atmosphere of the projector shall conform to the requirements of 3.12.13.

4.6.7 Periscope: XM50.

4.6.7.1 Orientation, optical characteristics.- The following tests shall be performed with the periscope positioned on a holding device, that permits the periscope to be mounted as required in 3.13.1 and pivoted in the horizontal plane. Position the back face of the coupling 67° 30' with respect to the horizontal mounting surface of the periscope. Unless

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otherwise specified the holding device shall be immobilized in a zero azimuth position. The image tube shall be removed from the periscope, and the clear filter shall be positioned in the optical train.

NOTE: The test values for all characteristics shall be recorded and made a part of the inspection record.

4.6.7.1.1 Contrast transfer.- Perform this test in accordance with the procedures specified in 4.6.5.1.1, 4.6.5.1.2 and 4.6.5.1.3, utilizing the testing equipment specified therein. See Figures 16 and 17 for orientation of the test equipment. The periscope shall meet to the applicable requirements of 3.11.3.1.1 and 3.11.3.1.2 to determine overall compliance with 3.13.1(e).

4.6.7.1.2 Radiant transmittance.- Perform this test in accordance with the procedures specified in 4.6.5.1.4, 4.6.5.1.5, 4.6.5.1.6 and 4.6.5.1.7, utilizing the testing equipment specified therein. See Figure 18 for orientation of the test equipment. The periscope shall meet the applicable requirements of 3.11.3.2.1a, 3.11.3.2.1b, 3.11.3.2.2a and 3.11.3.2.2b to determine overall compliance with 3.13.1f.

4.6.7.1.3 Resolution.- Perform this test in accordance with the procedures specified in 4.6.5.1.9, 4.6.5.1.10, 4.6.5.1.11 and 4.6.5.1.12, utilizing the testing equipment specified therein. See Figure 19 for orientation of the test equipment. The periscope shall meet the applicable requirements of 3.11.3.4.1a, 3.11.3.4.1b, 3.11.3.4.1.1a and 3.11.3.4.1b, to determine overall compliance with 3.13.1g.

4.6.7.1.4 Periscope with image tube installed.- Unless otherwise specified, the following tests shall be performed with the image tube and the reticle projector energized. The voltage required to energize the image tube and reticle projector shall be supplied from a power source capable of supplying and maintaining the required voltage. The image tube shall be permitted a stabilization period of 15 seconds prior to conducting these tests.

4.6.7.1.5 Field of view.- Perform this test in accordance with the procedures specified in 4.6.5.1.22. The field of view shall not be less than the value specified in 3.11.3.9 to determine overall compliance with 3.13.1h.

4.6.7.1.6 Image steadiness.- Perform this test in accordance with the procedures specified in 4.6.5.1.25 and 4.6.5.1.26. The periscope shall meet the applicable requirements of 3.11.3.11.1 and 3.11.3.11.2 to determine overall compliance with 3.13.1i.

4.6.7.2 Orientation optical accuracies.- The following tests shall be conducted with the periscope and appropriate equipment positioned as specified in 3.13.1 on special testing equipment conforming to 4.5.3.4.4. The viewing autocollimator shall be autocollimated to Datum Plane "A" shown on Sheet 1 of Drawing F10539736. The optical axis of the target projector collimator shall be alined parallel with respect to the optical

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axis of the viewing autocollimator and autocollimated to Datum Plane "Y" (see Sheet 1 of Drawing F10539736). The reticle pattern of the target collimator shall be projected by means of diffuse luminance from a 2870°K tungsten lamp. Luminance shall not exceed 1 x 10⁻1 foot-lamberts. Autocollimation and alinement shall be accomplished with the periscope removed from the line of sight through the collimators. The image tube and reticle projector shall be energized (see 4.6.7.1.5).

4.6.7.2.1 Excursion.- Perform this test in accordance with the procedures specified in 4.6.4.1.2 utilizing the auxiliary testing equipment specified therein. The periscope shall meet the requirements specified in 3.10.2 to determine overall compliance with 3.13.1a.

4.6.7.2.2 Elevation accuracy.- Perform this test in accordance with the procedures specified in 4.6.4.1.3. The periscope shall meet the requirements of 3.10.3 to determine overall compliance with 3.13.1b.

4.6.7.2.3 Backlash.- Perform this test in accordance with the procedures specified in 4.6.4.1.5. The periscope shall meet the requirements of 3.10.3 to determine overall compliance with 3.13.1c.

4.6.7.2.4 Mirror repeatability.- Perform this test in accordance with the procedures specified in 4.6.5.2.1. The periscope shall meet the requirements of 3.11.1.2 to determine overall compliance with 3.13.1d.

4.6.7.2.5 Horizontal excursion.- Perform this test in accordance with the procedures specified in 4.6.6.1.2. The periscope shall meet the requirements of 3.12.2.1 to determine overall compliance with 3.13.1j.

4.6.7.2.6 Vertical excursion.- Perform this test in accordance with the procedures specified in 4.6.6.1.3. The periscope shall meet the requirements of 3.12.2.2 to determine overall compliance with 3.13.1k.

4.6.7.2.7 Horizontal reticle travel.- Perform this test in accordance with the procedures specified in 4.6.6.1.4. The periscope shall meet the requirements of 3.12.2.3 to determine overall compliance with 3.13.1L.

4.6.7.2.8 Vertical reticle travel.- Perform this test in accordance with the procedures specified in 4.6.6.1.5. The periscope shall meet the requirements of 3.12.2.4 to determine overall compliance with 3.13.1m.

4.6.7.2.9 Parallax, daylight system.- This test shall be performed with the diopter knob of the periscope set for best focus of the periscope reticle. A line of sight through the reticle in the periscope shall be directed to a calibrated target located at 1000 meters ± 100 meters distant. Parallax is any apparent movement of the reticle lines in relation to the target image when the observer's head is moved from side to side or up and down. Parallax at the center of the field of view shall not exceed the tolerance specified in 3.13.2.

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4.6.7.2.10 Parallelism of reticle image.- Direct a line of sight through the eyepiece of the periscope and obtain coincidence of one end of the periscope's horizontal reticle line with that of the horizontal target line. Then observe for coincidence at the opposite end of the periscope's horizontal reticle line. The horizontal reticle line of the periscope shall be parallel to the horizontal target line image within the tolerance specified in 3.13.3.

4.6.7.2.11 Repeatability.- Observe through the eyepiece of the periscope assembly, establish coincidence of the reticle boresight mark with a point on the projected target image. Record the position of the boresight mark with respect to the target image. Repeated actuation of the introduction lever through the excursion specified in 3.13.4, for a minimum of three times, shall be performed. Actuation of the lever as specified in 3.13.4 shall not cause any shift in the image in excess of the limit specified in 3.13.4 as measured on the target image.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10539750. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Tank, Periscope XM50 is intended for use as the gunner's sighting and observation system in the Gun, Tank: M60A1E1.

6.2 Ordering data.- Procurement documents should specify:

a. Title, date, and number of this specification.
b. Order of document precedence.
c. Applicable packaging data sheet number.
d. Selection of applicable levels of preservation, and packing.
e. Availability of classified documents (see 3.11.3.5).
f. Submission of test data by the contractor.
g. Contractor's handling of classified test data.

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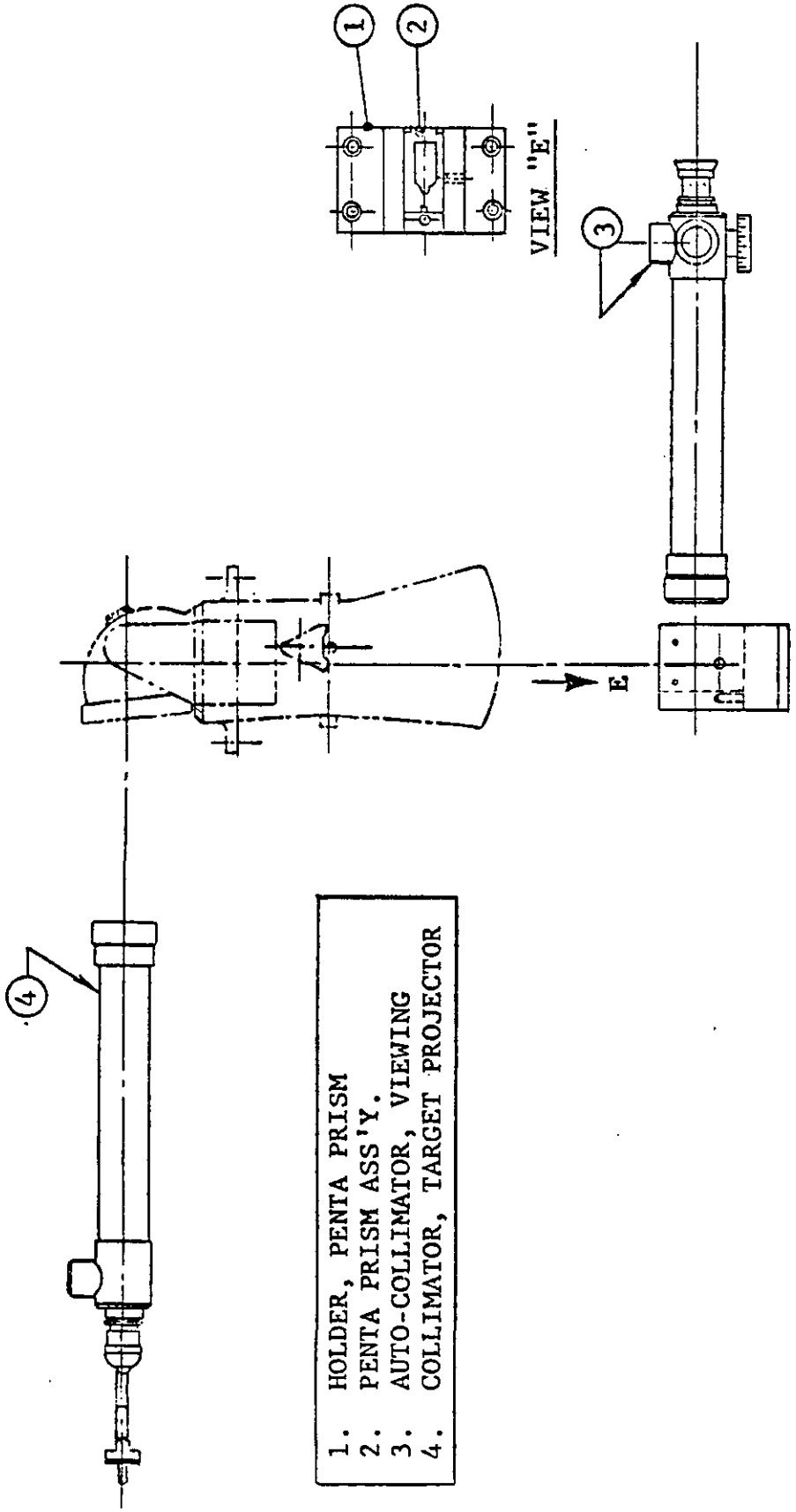


FIGURE 1

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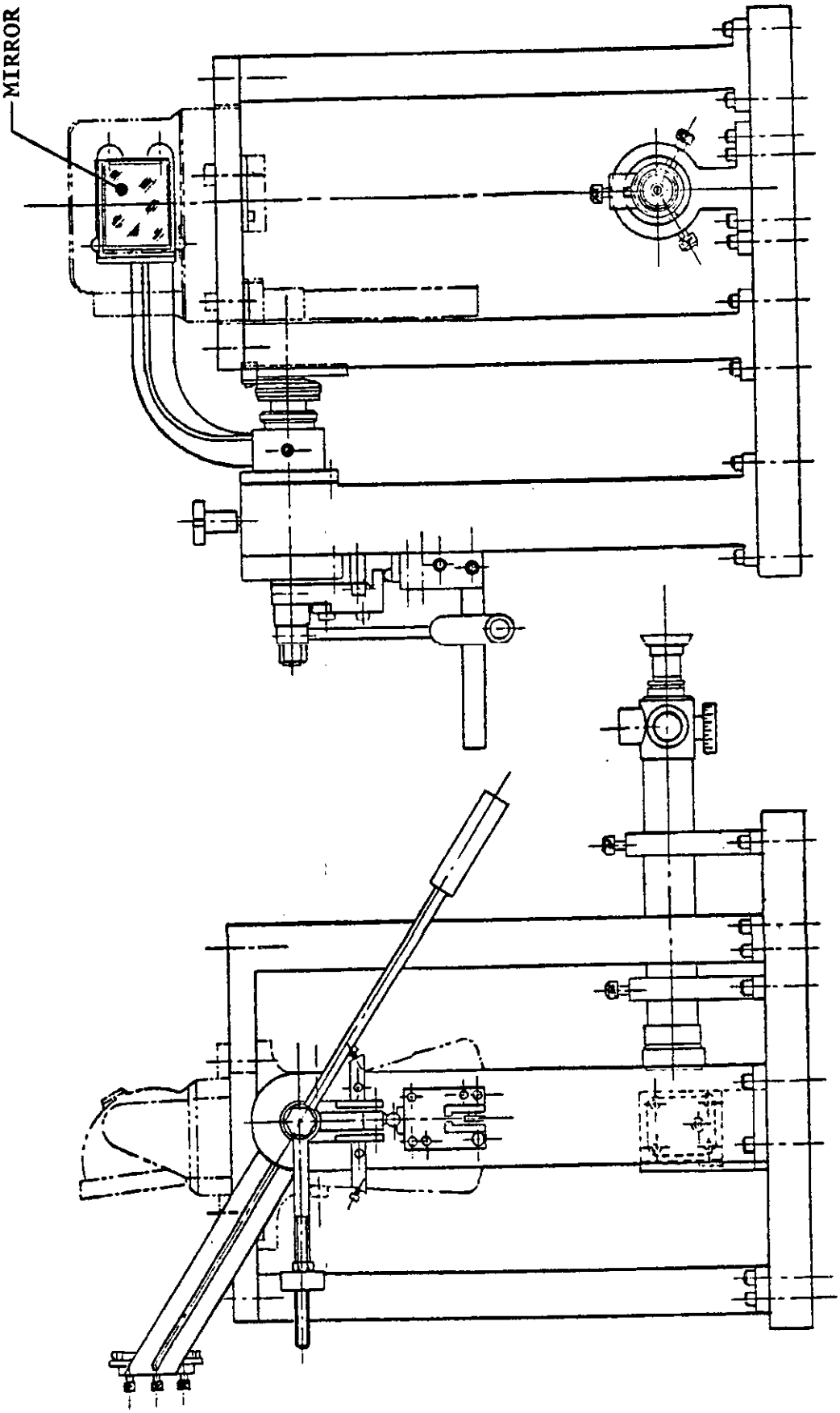
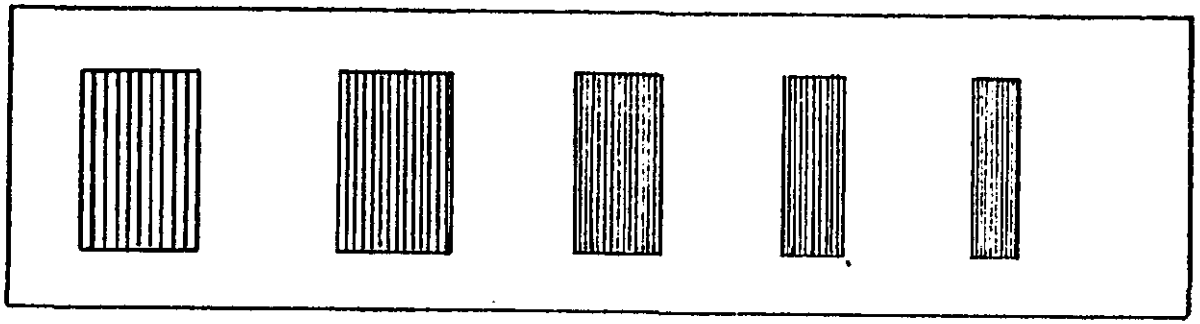


FIGURE 2

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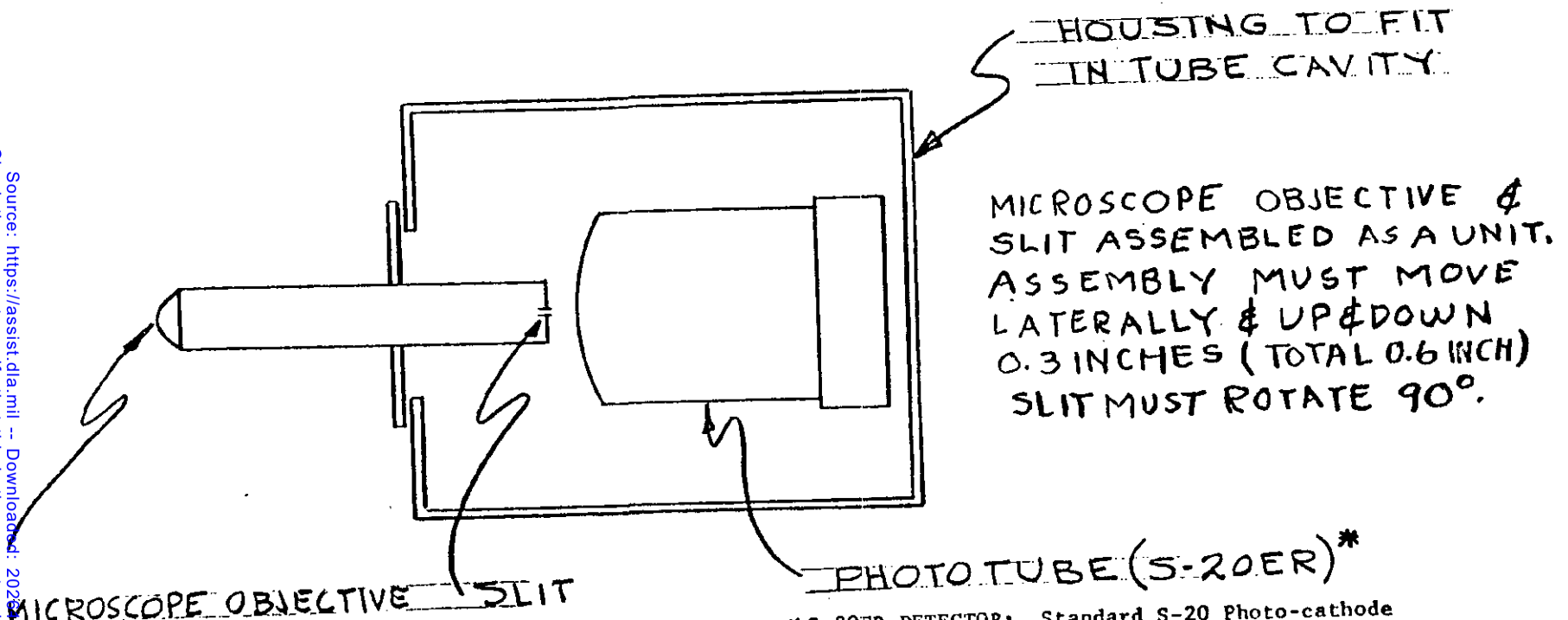
EACH TARGET GROUP WILL HAVE NOT LESS THAN 7 LINE PAIRS OF A PATTERN WHICH WILL GIVE THE INDICATED ANGULAR SUBTENSE

"A": 0.58 LINE PAIR/MILLI-RADIAN - 2.5 L.P./MI AT P.C. OF IMAGE INTENSIFIER.
"B": 1.73 LINE PAIR/MILLI-RADIAN - 7.5 L.P/MI AT P.C. OF IMAGE INTENSIFIER.
"C": 4.62 LINE PAIR/MILLI-RADIAN - 20 L.P/MI AT P.C. OF IMAGE INTENSIFIER.
"D": 6.92 LINE PAIR/MILLI-RADIAN - 30 L.P/MI AT P.C. OF IMAGE INTENSIFIER.
"E": 9.23 LINE PAIR/MILLI-RADIAN - 40 L.P./MI AT P.C. OF IMAGE INTENSIFIER.

FREQUENCY PATTERN - CONTRAST TRANSFER

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
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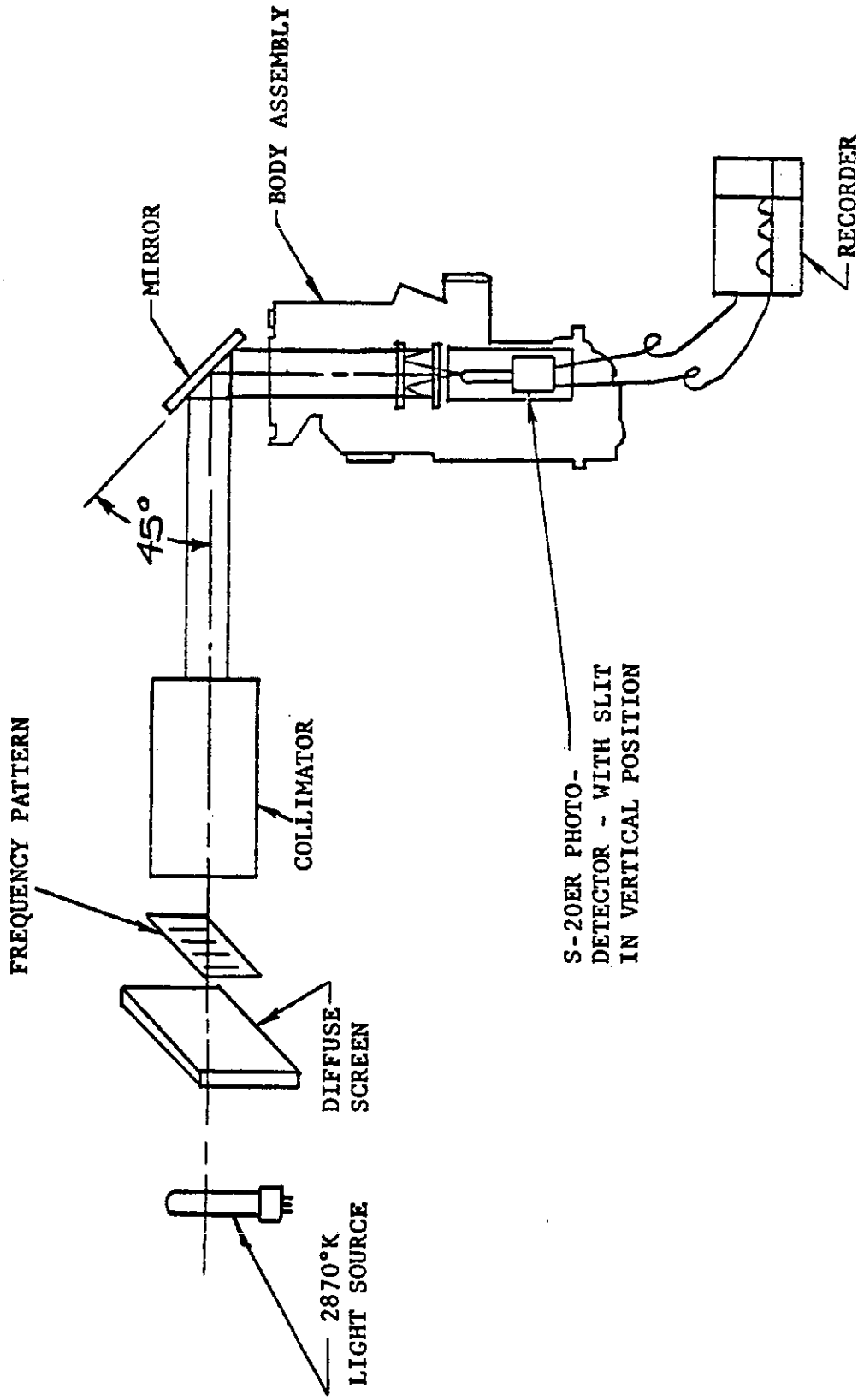


*S-20ER DETECTOR: Standard S-20 Photo-cathode response extended at the red end so that the sensitivity at 0.8 micron and at 0.85 are 85% of the minimum values in the Purchase Description for the Image Intensifier Tube.

S-20ER DETECTOR AND SLIT

FIGURE 4

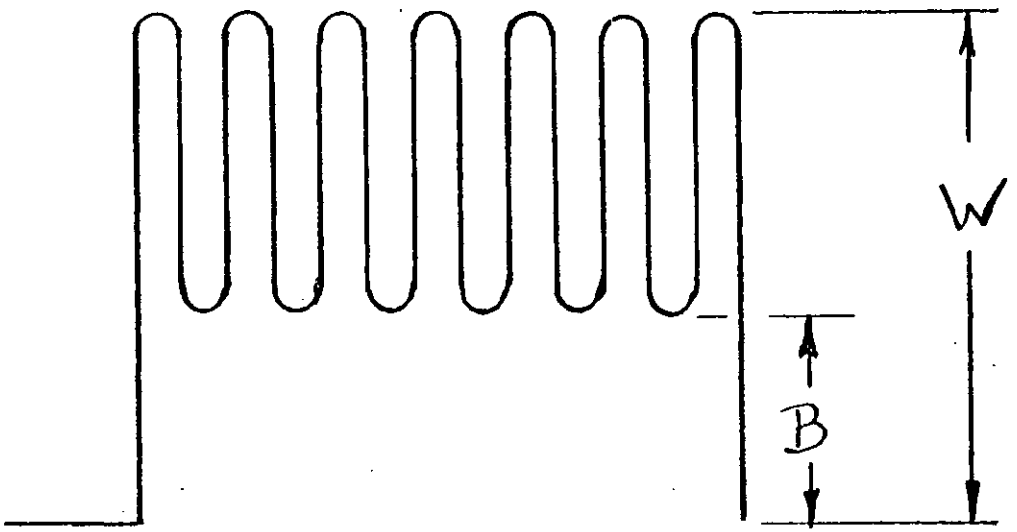
Source: https://assist.dla.mil -- Downloaded: 2025-10-26T04:20Z
Check the source to verify that this is the current version before use.



CONTRAST TRANSFER
BODY ASSEMBLY OBJECTIVE

FIGURE 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
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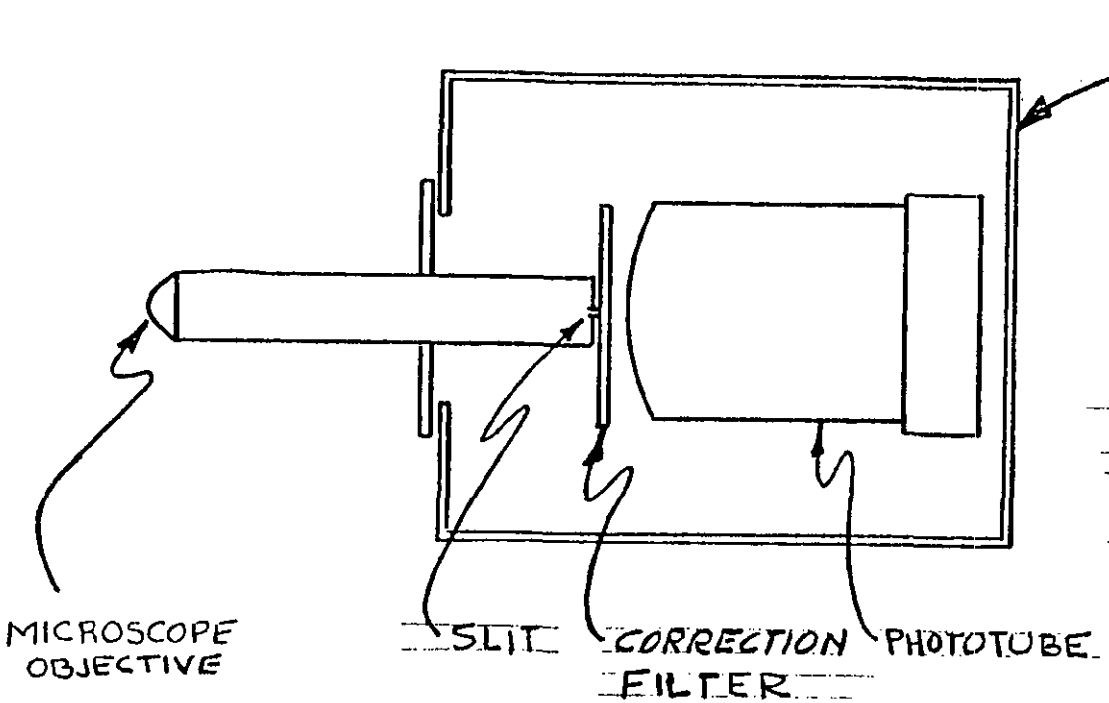


NOTE: CONTRAST TRANSFER IS DEFINED AS $\frac{W-B}{W+B}$

CONTRAST TRANSFER WAVE FORM

FIGURE 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



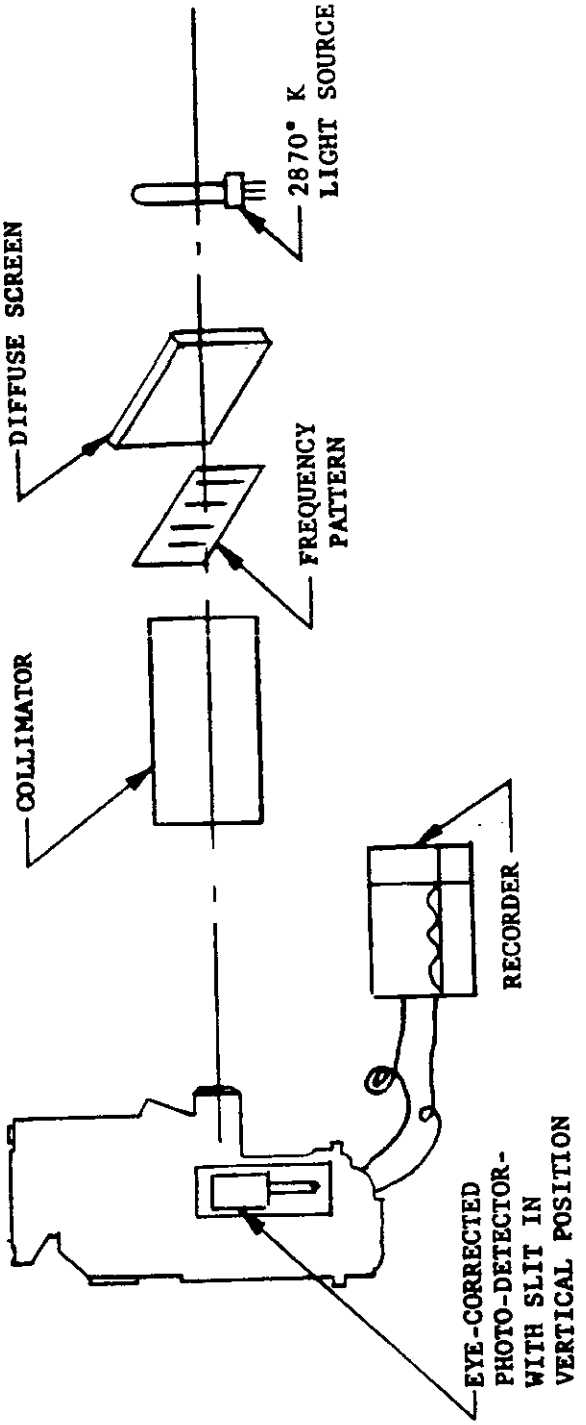
MICROSCOPE OBJECTIVE & SLIT ASSEMBLED AS A UNIT. ASSEMBLY MUST MOVE LATERALLY & UP & DOWN 0.3 INCHES (TOTAL 0.6 INCH) SLIT MUST ROTATE 90°.

CORRECTION FILTER WILL CHANGE TUBE RESPONSE TO STANDARD VISIBILITY FUNCTION (V_2).

EYE-CORRECTED DETECTOR AND SLIT

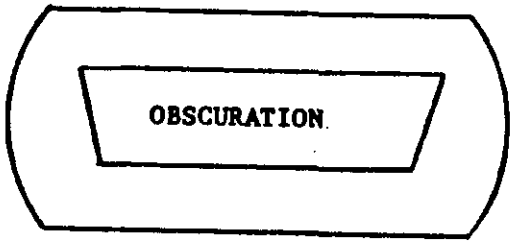
FIGURE 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

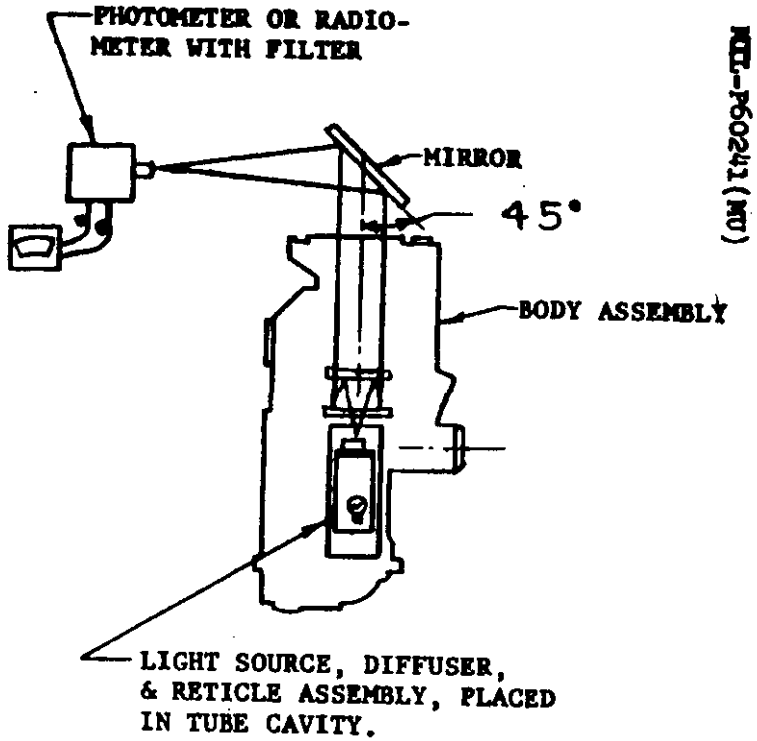


CONTRAST TRANSFER
BODY ASSEMBLY EYEPIECE
FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



PERISCOPE ENTRANCE PUPIL AS SEEN FROM IN FRONT OF HEAD.

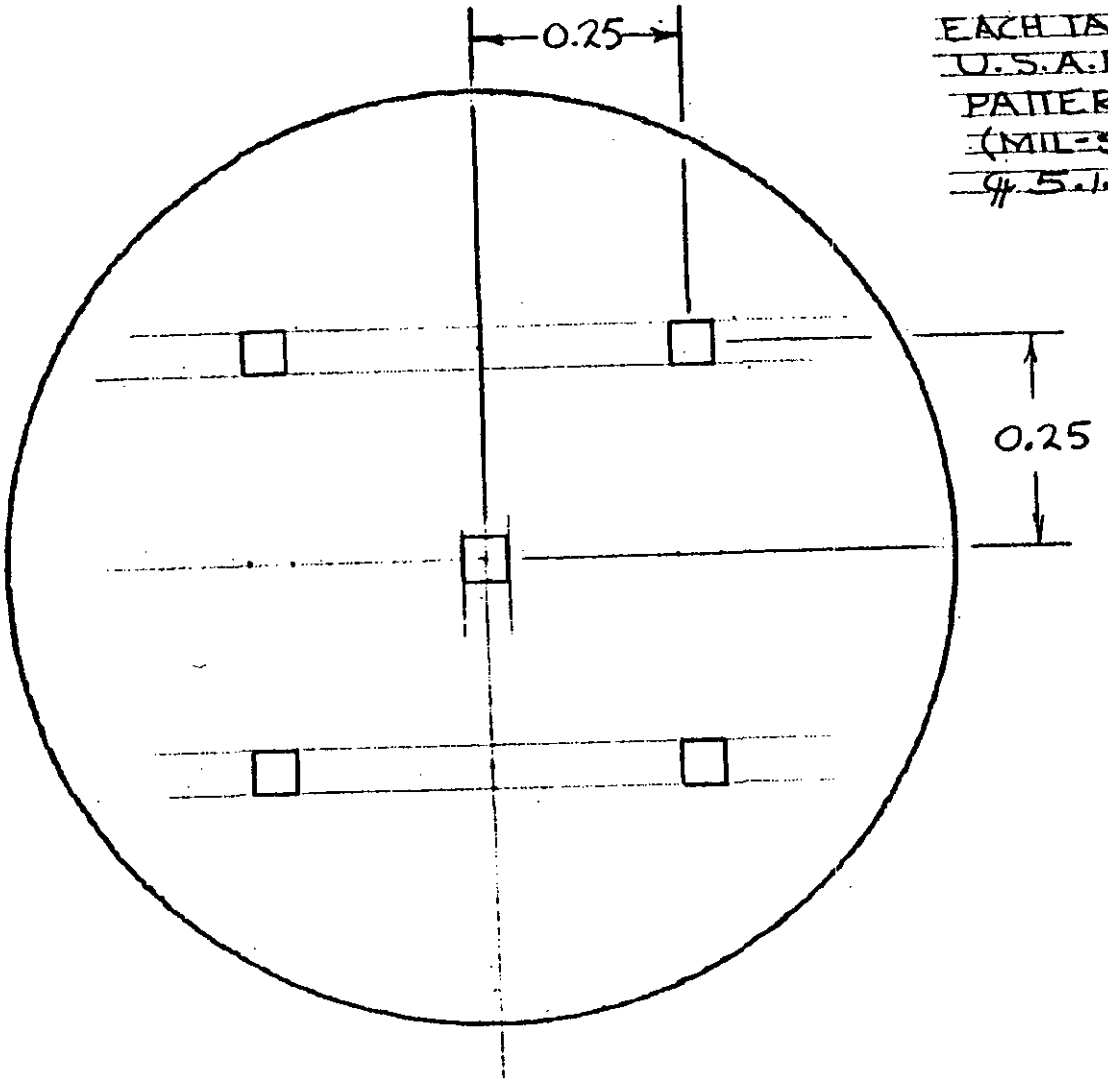


MIL-P-60241(MU)

OBJECTIVE TRANSMISSION, VISIBLE & I. R.

FIGURE 9

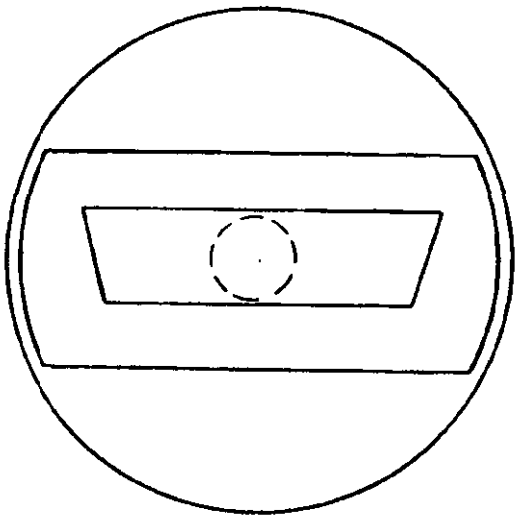
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



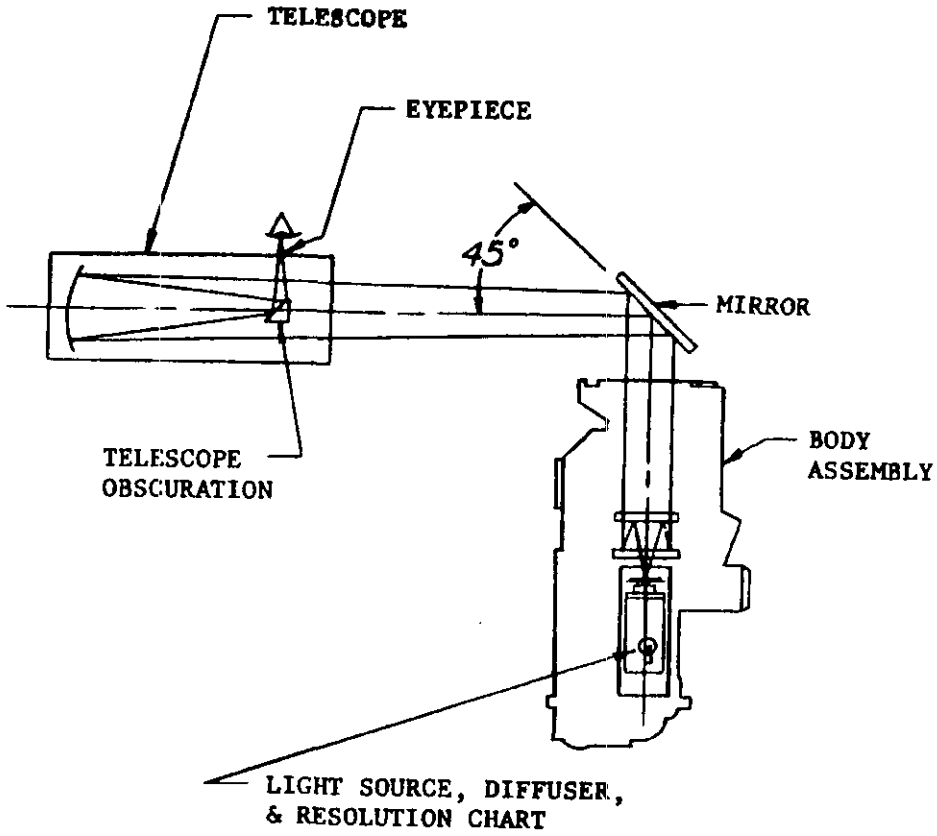
EACH TARGET TO BE A
U.S.A.F. 1951 RESOLUTION
PATTERN UP TO 228 L.P./MM
(MIL-STD-150, 12 MAY 1959)
5.1.1.7

RETICLE PATTERN
FOR
OBJECTIVE RESOLUTION TEST
FIGURE 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



CARE MUST BE TAKEN TO SUPERIMPOSE THE OBSCURATIONS OF THE PERISCOPE AND THE TELESCOPE

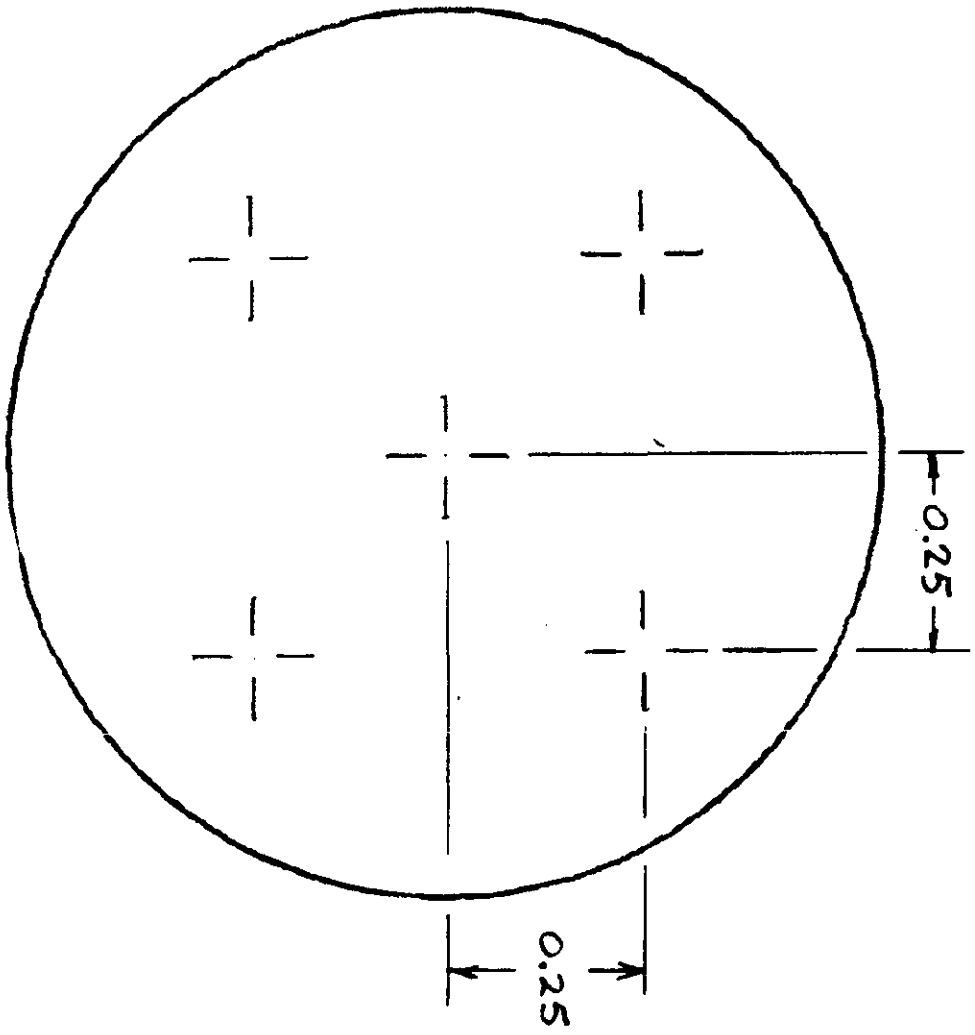


RESOLUTION OBJECTIVE ASSEMBLY

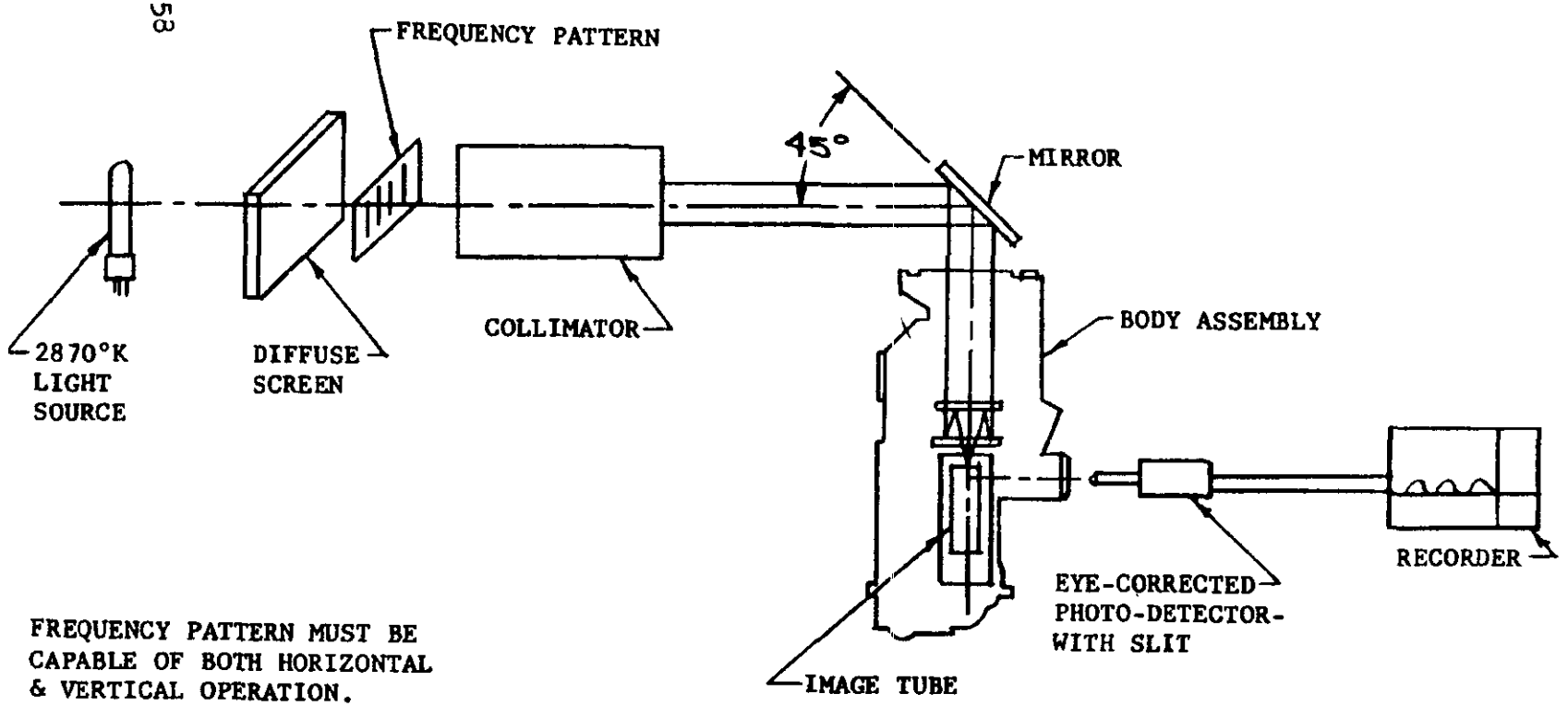
FIGURE 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.

RETICLE PATTERN
FOR
T/# TEST
FIGURE 12



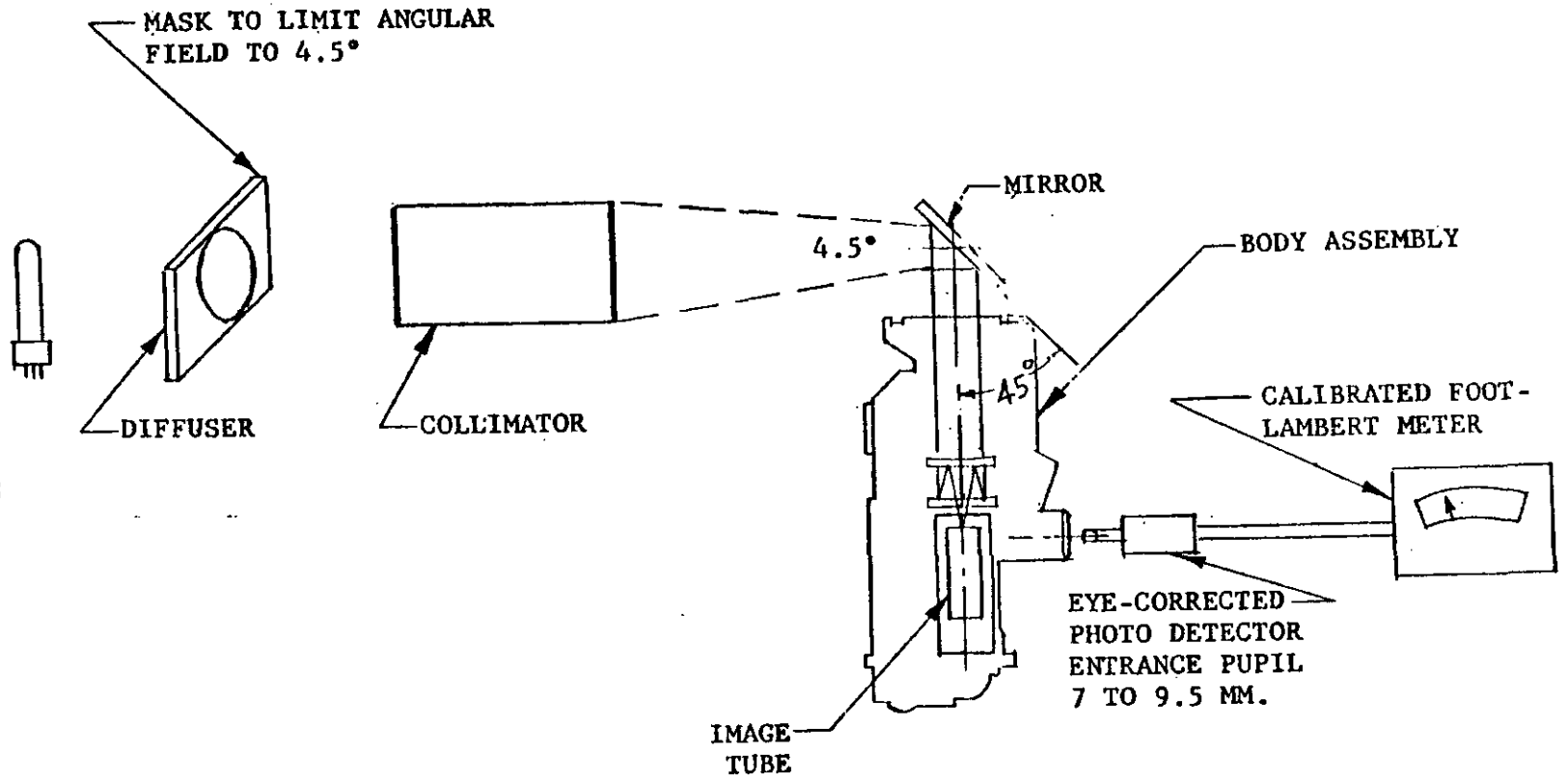
Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:20Z
Check the source to verify that this is the current version before use.



CONTRAST TRANSFER
BODY ASSEMBLY

FIGURE 13

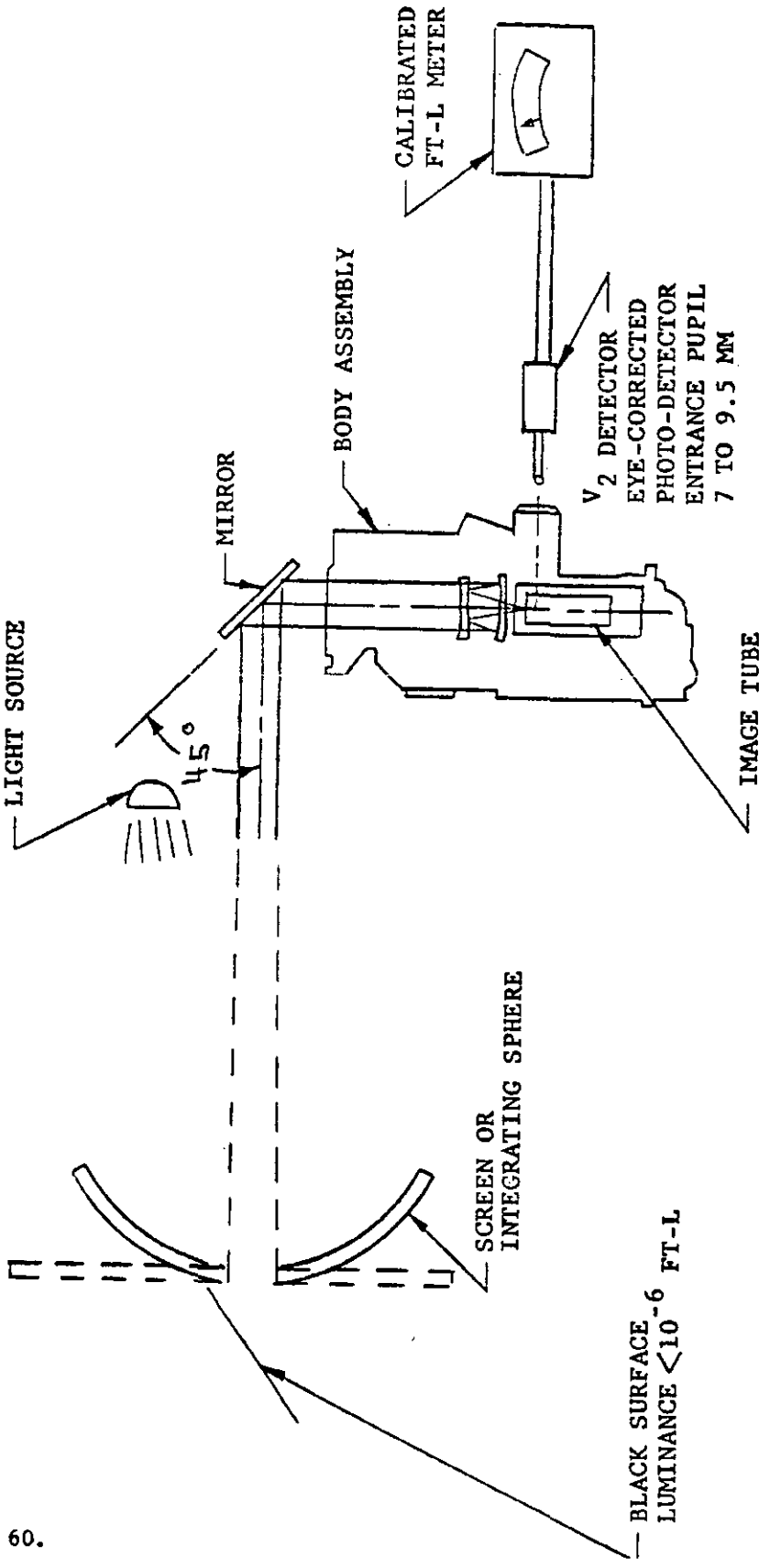
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



LUMINANCE GAIN

FIGURE 14

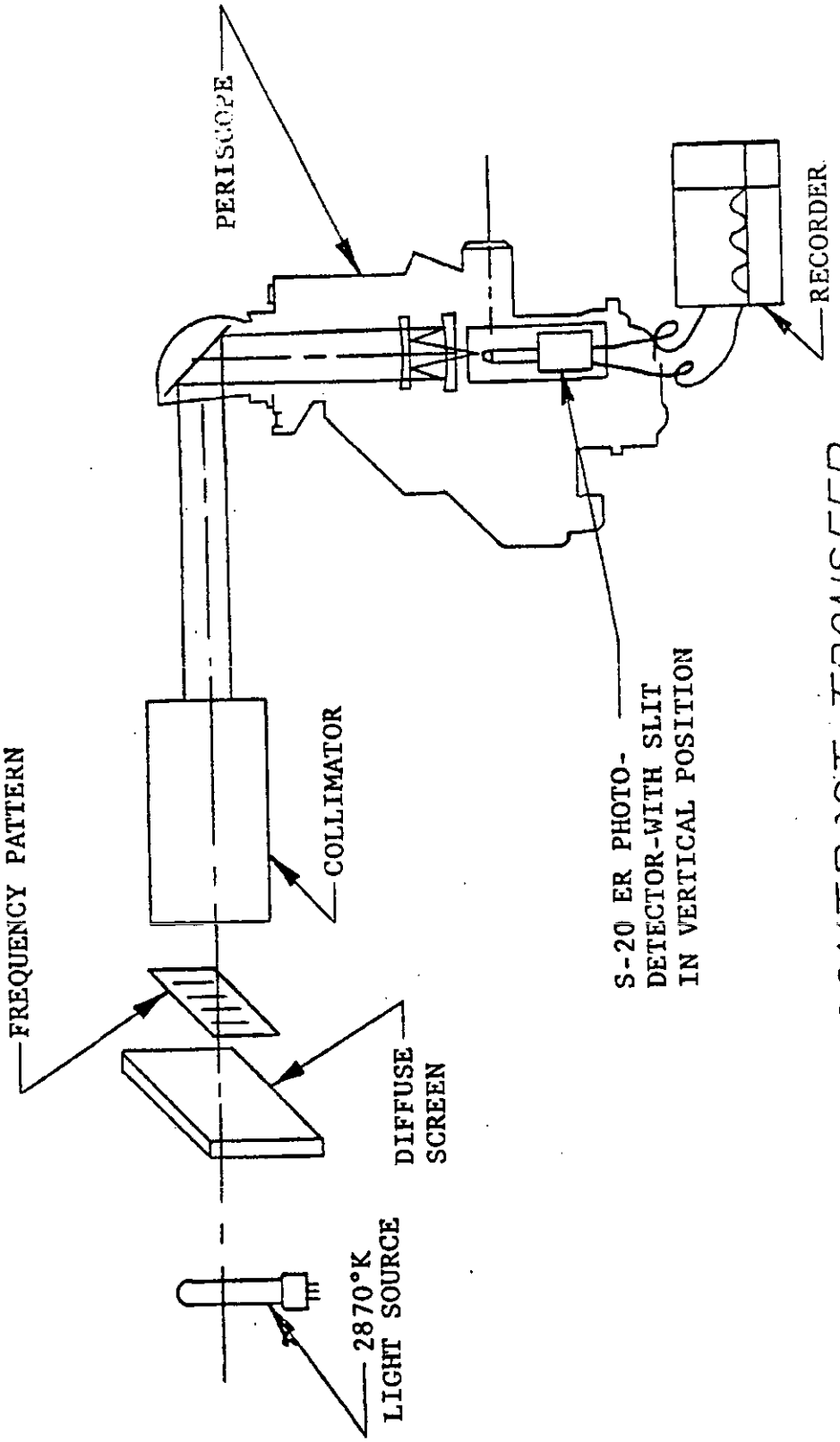
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
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STRAY-LIGHT MEASUREMENT

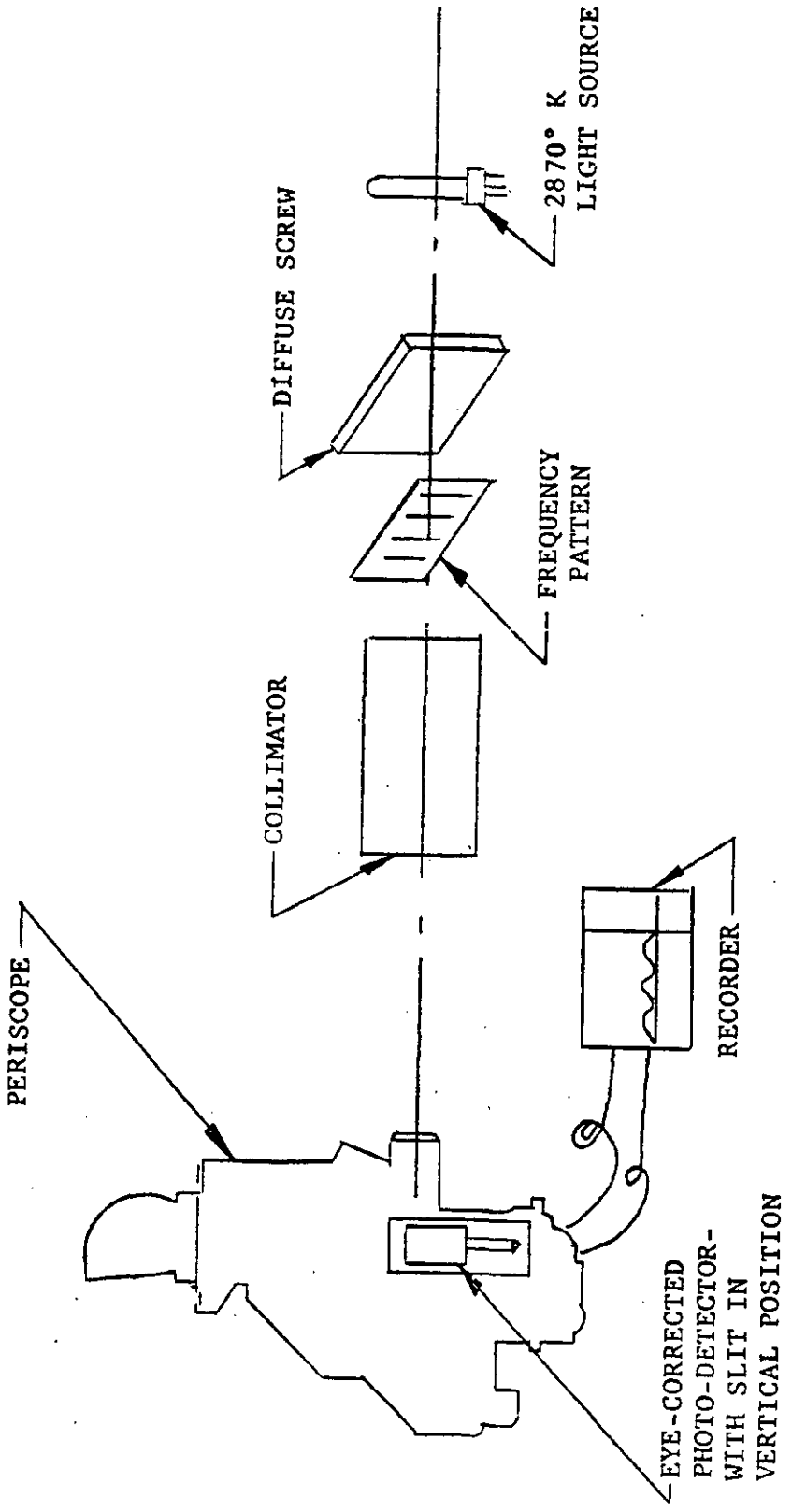
FIGURE 15

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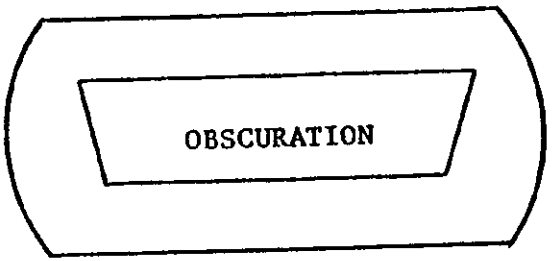
CONTRAST TRANSFER
PERISCOPE OBJECTIVE
FIGURE 16.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
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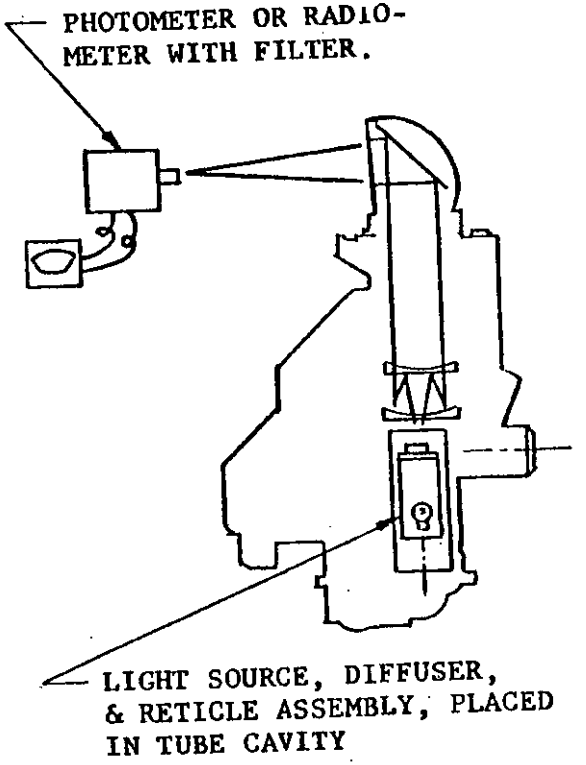


CONTRAST TRANSFER
PERISCOPE EYEPIECE
FIGURE 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:20Z
Check the source to verify that this is the current version before use.



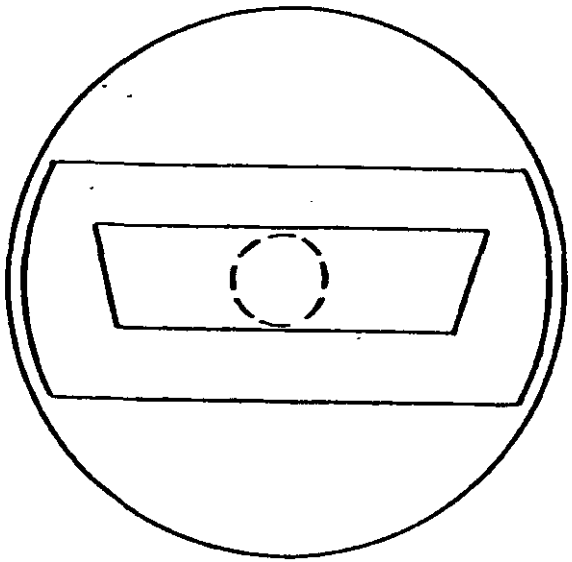
PERISCOPE ENTRANCE PUPIL AS SEEN FROM IN FRONT OF HEAD.



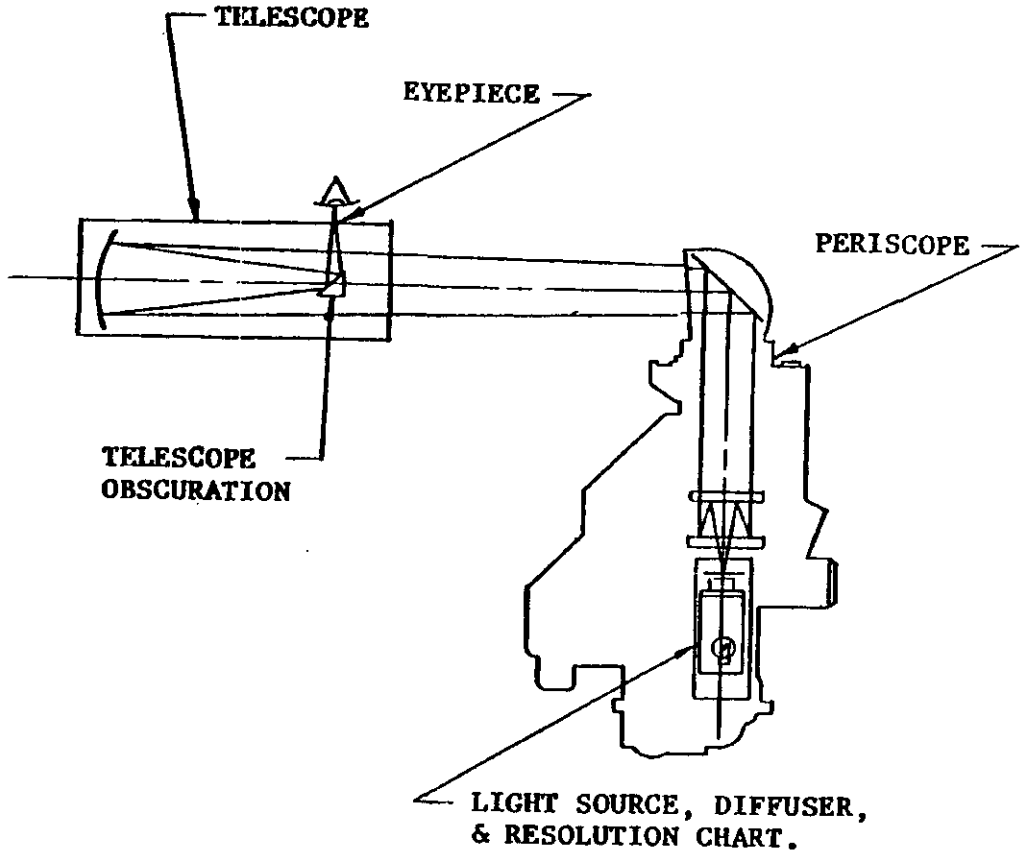
OBJECTIVE TRANSMISSION, VISIBLE & I.R.

FIGURE 18

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CARE MUST BE TAKEN TO SUPERIMPOSE THE OBSCURATIONS OF THE PERISCOPE AND THE TELESCOPE.



RESOLUTION OBJECTIVE ASSEMBLY

FIGURE 19

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FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
FRANKFORD ARSENAL
ATTN: SMUFA-N1160
PHILA, PA. 19137

FOLD

PLATE NO. 15419 (BAC)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 18419

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